

The Same Education, Different Intimate Partner Violence Attitude Gradients

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Presentation Outline

- 1 Introduction
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- 6 Conclusion

Motivation

- Education is widely expected to change gender-related attitudes, including attitudes toward intimate partner violence.
- Women with lower-secondary education are, on average, less likely to justify intimate partner violence, but there has not yet been comparable empirical evidence.
- The education gradient varies sharply across countries—from strongly negative to nearly zero.
- Under what social conditions does women's education translates into lower acceptance of domestic violence (or IPV) across countries?
- This paper asks whether country-level male and female secondary-school attainment moderates the effect of individual education on attitudes toward domestic violence.

This Paper

- Research question:
 - How much does the association between women's education and IPV acceptance vary across countries?
 - Does male lower-secondary completion predict a stronger negative female education gradient?
- Model: A norm-dependent attitude-expression model
 - Female education shifts private beliefs and relative norm sensitivity, while male education attainment ratio raises the IPV-related intolerant norm, represented as the level of women's IPV opposition that can be expressed without normative sanction.

This Paper

- Empirical Setting: Multiple Indicator Cluster Survey (MICS) 2005–2024, linked with country-level World bank data
 - Individuals aged 25–49 across 74 female-sample countries and 42 male-sample countries, with harmonized measures of IPV attitudes and lower-secondary attainment.
- Empirical Strategy: Two-step cross-country design
 - First stage: estimate country-specific education gradients in IPV acceptance, identified under selection on observables approach.
 - Second stage: explain their variation using male and female secondary-school attainment, macro controls.

Preview of Main Findings

- **First stage mostly negative, but highly heterogeneous.**
 - 58 of 73 country-specific female education gradients are significantly negative, but their magnitudes vary.
- **Male education attainment rate steepens the gradient; female attainment flattens it.**
 - A 10-percentage-point increase in male secondary+ attainment is associated with a female education gradient that is 1.8 percentage points (34.8 percent of baseline mean) more negative, whereas the same increase in female attainment is associated with a gradient that is 2.2 percentage points (42.6 percent of baseline mean) less negative.
- **Little evidence emerges for men.**
 - The corresponding cross-country relationships are small and imprecisely estimated in the male sample.

Contribution and Related Literature

1. Comparable country-specific education gradients in IPV acceptance.

- Prior studies focus on the effect within country analysis (Friedman et al. 2016; Bhuwania and Heymann 2022; Dhar, Jain, and Jayachandran 2022).
- Using harmonized MICS data, I estimate country-specific education gradients—under selection on observables approach— for women in 73 countries (men in 42 countries).

2. Cross-country evidence that the female education gradient is context-dependent.

- Prior within-country evidence shows that women's IPV attitudes respond to neighbourhood norms (Dhamija et al. 2024). I show that the gradient is more negative where male secondary+ attainment is higher, but less negative where female secondary+ attainment is higher. The corresponding estimates for men are small and imprecise.

3. An education-dependent extension of norm-based utility models.

- Existing norm- and fairness-based models generally treat the relevant norm as given within a context (Cappelen et al. 2007; López-Pérez 2008; Krupka and Weber 2013; Kimbrough and Vostroknutov 2016).
- I allow own education to shift private beliefs and relative sensitivity to normative sanctions, while country-level educational attainment shapes the normative environment.

1. Setup: Male Normative Environment

Woman j in country c chooses an expressed IPV-intolerant attitude:

$$A_{jc} \in [0, 1], \quad A_{jc} \uparrow \Rightarrow \text{stronger expressed opposition to IPV.}$$

Private belief

$$B_{jc}^F = B^F(e_{jc}^F), \quad B^{F'}(e_{jc}^F) > 0.$$

Female education shifts private beliefs toward stronger opposition to IPV.

Normative environment

$$S_c = S(E_c^M), \quad \frac{\partial S_c}{\partial E_c^M} > 0.$$

Male educational attainment is used as a proxy for the normative environment for IPV.

Sanction-free threshold

S_c is the IPV-related intolerant norm, represented as the level of women's IPV opposition that can be expressed without normative sanction.

$$A_{jc} \leq S_c \Rightarrow \text{no sanction}, \quad A_{jc} > S_c \Rightarrow \text{sanction cost.}$$

2. Education-Dependent Sensitivity

The woman faces two expression costs:

- private-belief mismatch: $A_{jc} \neq B^F(e_{jc}^F)$;
- normative sanction when $A_{jc} > S_c$.

Allow education to affect both sensitivity parameters:

$$\pi_j = \pi(e_{jc}^F) > 0, \quad \phi_j = \phi(e_{jc}^F) > 0.$$

The original problem is

$$A_{jc}^* = \arg \min_{A_{jc} \in [0,1]} \left[\pi(e_{jc}^F) h(A_{jc} - B^F(e_{jc}^F)) + \phi(e_{jc}^F) g((A_{jc} - S_c)_+) \right], \quad h', h'', g', g'' > 0.$$

Since only the relative weight matters,

$$A_{jc}^* = \arg \min_{A_{jc} \in [0,1]} \left[h(A_{jc} - B^F(e_{jc}^F)) + \rho(e_{jc}^F) g((A_{jc} - S_c)_+) \right], \quad \rho(e_{jc}^F) \equiv \frac{\phi(e_{jc}^F)}{\pi(e_{jc}^F)} > 0.$$

3. Two Cases

Case 1: Non-binding sanction

$$B^F(e_{jc}^F) \leq S_c.$$

Then

$$A_{jc}^* = B^F(e_{jc}^F).$$

So

$$\frac{\partial A_{jc}^*}{\partial e_{jc}^F} = B^{F'}(e_{jc}^F) > 0,$$

$$\frac{\partial^2 A_{jc}^*}{\partial e_{jc}^F \partial S_c} = 0.$$

Case 2: Binding sanction

$$B^F(e_{jc}^F) > S_c,$$

$$S_c < A_{jc}^* < B^F(e_{jc}^F).$$

Expression is shaped by both:

- private beliefs $B^F(e_{jc}^F)$;
- relative norm sensitivity $\rho(e_{jc}^F)$.

This is the main case below.

4. First-Order Condition

In the binding region, define

$$x_{jc} = A_{jc}^* - B^F(e_{jc}^F) < 0, \quad z_{jc} = A_{jc}^* - S_c > 0.$$

The first-order condition is

$$h'(x_{jc}) + \rho(e_{jc}^F)g'(z_{jc}) = 0.$$

The second-order condition is

$$D_{jc} \equiv h''(x_{jc}) + \rho(e_{jc}^F)g''(z_{jc}) > 0.$$

Interpretation

The expressed attitude balances private-belief consistency against normative-sanction costs.

5. Female Education Effect

Differentiating the FOC with respect to female education:

$$\frac{\partial A_{jc}^*}{\partial e_{jc}^F} = \frac{h''(x_{jc})B^{F'}(e_{jc}^F) - \rho'(e_{jc}^F)g'(z_{jc})}{D_{jc}}.$$

The education effect has two channels:

$$\underbrace{\frac{h''(x_{jc})B^{F'}(e_{jc}^F)}{D_{jc}}}_{\text{private-belief channel}} + \underbrace{\frac{-\rho'(e_{jc}^F)g'(z_{jc})}{D_{jc}}}_{\text{norm-sensitivity channel}}.$$

In the assumption,

$$B^{F'}(e_{jc}^F) > 0.$$

Education shifts private beliefs toward IPV opposition. The sign and magnitude of the female education effect depend on the combined contribution of the private-belief and norm-sensitivity channels.

6. Moderation Effect and Its Components

Private-belief mismatch channel

$$A_{BS} \equiv \frac{\partial^2 A_{jc}^*}{\partial B_{jc}^F \partial S_c}$$

$$= \frac{\rho[\{h''(x_{jc})\}^2 g'''(z_{jc}) + \rho\{g''(z_{jc})\}^2 h'''(x_{jc})]}{D_{jc}^3}.$$

A_{BS} measures how a more supportive norm changes the transmission of private beliefs into expressed attitudes.

Since $S_c = S(E_c^M)$,

$$\frac{\partial^2 A_{jc}^*}{\partial e_{jc}^F \partial E_c^M} = \left[B^{F'}(e_{jc}^F) A_{BS} + \rho'(e_{jc}^F) A_{\rho S} \right] \frac{\partial S_c}{\partial E_c^M}.$$

Normative sanction channel

$$A_{\rho S} \equiv \frac{\partial^2 A_{jc}^*}{\partial \rho \partial S_c}$$

$$= \frac{h''(x_{jc}) g''(z_{jc}) D_{jc} + \rho g'(z_{jc}) [h'''(x_{jc}) g''(z_{jc}) - h''(x_{jc}) g'''(z_{jc})]}{D_{jc}^3}.$$

$A_{\rho S}$ measures how a more supportive norm changes the effect of relative sanction sensitivity on expressed attitudes.

7. Sign Prediction

- $A_{BS} > 0$: The more the environment is toward expressions opposing IPV, the more strongly the private beliefs shaped by education are reflected in expressed attitudes (Dhamija et al. 2024).
- $A_{\rho S} > 0$: The more the women is sensitive to social norms, the more strongly they react to opposition in the norm environment.
- $\rho' \geq 0$: Male education unambiguously strengthens the female education effect.
- $\rho' < 0$: The sign depends on whether

$$B^{F'} A_{BS} \begin{matrix} \geq \\ \leq \end{matrix} |\rho'| A_{\rho S}.$$

Main Clarifications to Support the Model

Main implications

- 1 $B^{F'}(e_{jc}^F) > 0$: Women's education shifts private beliefs toward stronger opposition to IPV.
- 2 $\frac{\partial^2 A_{jc}^*}{\partial E_c^M \partial e_{jc}^F} > 0$: a more educated male environment strengthens the positive effect of women's education on expressed IPV opposition. Equivalently, the female education coefficient on IPV acceptance becomes more negative.

Further empirical checks

- 3 $\rho'(e_{jc}^F)$: If $\rho'(e_{jc}^F) < 0$, women's education reduces sensitivity to normative sanctions relative to sensitivity to private-belief mismatch.
- 4 $\frac{\partial S_c}{\partial E_c^M} > 0$: Male education completion share raises the normative threshold faced by women.

Data

Individual Level Data: Multiple Indicator Cluster Surveys (MICS) 2005 ~ 2023

- Cross sectional data from Unicef.
- Data availability is irregular across countries and survey rounds.

Macro Level Data: World Bank Data, World Population Review, OECD Social Institutions and Gender Index

Data: World Bank

- GDP per capita, employment rate, population, duration of education, Urban density (2005 ~ 2023)
- Education attainment share data of aged between 25 or above (2005 ~ 2023)

Data: World Population Review

- Major religion in country 2010, 2020

Data: OECD Social Institutions and Gender Index

- Patriarchal family institutions index 2019

Sample restriction in Individual Level Data

- Survey sample is different by country.
- IPV related questions are only surveyed from age 15 to 49.
- The sample is restricted to individuals aged 25–49 to match the data from educational attainment data in World Bank..
 - Female sample: **74 countries** (Tajikistan only married samples)
 - Male sample: **42 countries**
- 5 countries where macro level data is not available are also not used (Kiribati, Republic of South Sudan, Kosovo, Turks and Caicos Island, Tuvalu)

Sample restriction in Macro Level Data

- Missing values in educational attainment are imputed using two methods:
 - **Linear interpolation** within the observed range (82%).
 - **Nearest-value extrapolation** outside the observed range (18%).
- For countries observed in multiple survey years, annual variables are aggregated using **survey-year observation weights**.
 - Example: If Country A is observed in 2008 and 2015, GDP per capita is calculated as a weighted average of the 2008 and 2015 values.
- Missing values in religion variables are supplemented using government reports and other reliable external sources.

Empirical Strategy

First stage

$$Y_{isr}^c = \alpha + \tau^{Edu} Edu_{isr}^c + K_{isr}^c{}' \Gamma + \delta_r^c + \omega_s^c + \epsilon_{isr}^c$$

Second stage

$$\hat{\tau}_c^{Edu} = \alpha + \beta_M EduRatioMale_c + \beta_F EduRatioFemale_c + X_c' \gamma + \epsilon_c$$

First-stage variables

- Y_{isr}^c : attitude toward domestic violence
- Edu_{isr}^c : lower-secondary-or-above education
- K_{isr}^c : age, age squared, wealth quintiles, urban residence
- ω_s^c and δ_r^c : region and survey round/year fixed effects
- Survey individual weights are included

Second-stage variables

- $\hat{\tau}_c^{Edu}$: first-stage education effect in country c
- $EduRatioMale_c$: male lower-secondary-or-above share
- $EduRatioFemale_c$: female lower-secondary-or-above share
- X_c : log GDP per capita, employment rate, major religion, urban population density, youth share
- Countries are weighted by inverse first-stage variance

Construction of Domestic Violence Attitude Measure

Survey item

Respondents were asked whether a husband is justified in hitting or beating his wife in the following situations:

- going out without telling him
- neglecting the children
- arguing with him
- refusing to have sex with him
- burning the food

Coding method

- **1**: answered "Yes" to at least one item
- **0**: answered otherwise
- **Missing (.)**: did not answer any item

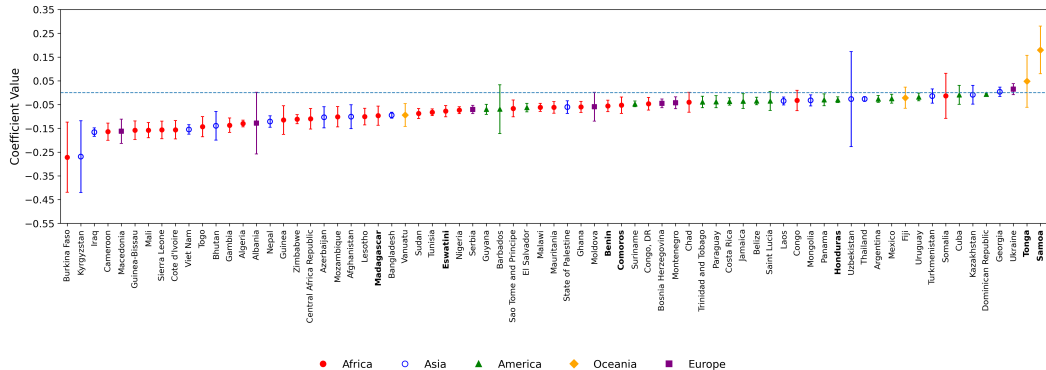
Summary Statistics

Variable	Female (73 countries)		Female (42 countries)		Male (42 countries)	
	Mean (SD)	N countries	Mean (SD)	N countries	Mean (SD)	N countries
Domestic Violence	0.276 (0.222)	73	0.298 (0.208)	42	0.202 (0.147)	42
Lower Secondary +	0.561 (0.266)	73	0.510 (0.261)	42	0.585 (0.219)	42
Age	35.298 (1.512)	73	35.173 (1.439)	42	35.572 (1.705)	42
Poorest	0.135 (0.043)	73	0.131 (0.040)	42	0.150 (0.040)	42
Second	0.164 (0.037)	73	0.163 (0.032)	42	0.175 (0.033)	42
Middle	0.186 (0.030)	73	0.187 (0.024)	42	0.193 (0.018)	42
Fourth	0.223 (0.021)	73	0.228 (0.019)	42	0.223 (0.024)	42
Richest	0.291 (0.101)	73	0.291 (0.077)	42	0.260 (0.061)	42
Urban	0.555 (0.175)	73	0.509 (0.159)	42	0.491 (0.167)	42
Currently Married	0.755 (0.101)	73	0.767 (0.074)	42	0.732 (0.120)	42

Mean of Specific IPV Attitudes

Variable	Mother (73 countries)	Mother (42 countries)	Father (42 countries)
Argues with husband	0.1488 (0.1533)	0.1526 (0.1401)	0.1048 (0.0910)
Burns food	0.0807 (0.0991)	0.0951 (0.1126)	0.0477 (0.0598)
Goes out without telling husband	0.1576 (0.1576)	0.1677 (0.1490)	0.0984 (0.0932)
Refuses sex	0.1244 (0.1472)	0.1340 (0.1440)	0.0710 (0.0729)
Neglects children	0.1976 (0.1668)	0.2172 (0.1626)	0.1297 (0.1066)
Notes: Entries are country-equal means. Parentheses report the standard deviation across country-specific weighted means. The first mother column uses all available mother countries; the second mother column is restricted to countries matched to the father sample.			

Female: Lower secondary + on IPV Norm



Coefficient Bounds under Unobservable Selection by Oster (2019)

Country	Restricted model				Controlled model				Oster approx. bounds		Sign-change BP			
	Estimate	95% CI	R^2	N	Estimate	95% CI	R^2	N	$\delta = 1$	$\delta = -1$	$+\infty$	$10 \hat{\beta} $	$3 \hat{\beta} $	$2 \hat{\beta} $
Madagascar	-0.137	[-0.200, -0.075]	0.089	2,884	-0.138	[-0.200, -0.077]	0.097	2,884	-0.141	-0.136	0.997	4.085	27.911	27.911
Comoros	-0.058	[-0.125, 0.008]	0.032	832	-0.058	[-0.124, 0.008]	0.038	832	-0.057	-0.059	1.000	2.474	8.091	9.223
Benin	-0.045	[-0.080, -0.011]	0.139	3,412	-0.041	[-0.076, -0.007]	0.146	3,412	-0.019	-0.064	0.528	0.528	0.528	0.528
Honduras	-0.023	[-0.037, -0.010]	0.034	8,934	-0.023	[-0.036, -0.010]	0.034	8,934	-0.019	-0.027	1.000	1.325	1.325	1.325
Eswatini	-0.017	[-0.055, 0.021]	0.023	943	-0.019	[-0.057, 0.019]	0.030	943	-0.021	-0.016	0.630	2.369	2.369	2.369
Tonga	0.050	[-0.084, 0.183]	0.078	1,202	0.048	[-0.085, 0.182]	0.083	1,202	0.042	0.055	1.000	1.725	1.725	1.725
Samoa	0.169	[0.029, 0.310]	0.049	1,419	0.166	[0.024, 0.307]	0.052	1,419	0.141	0.190	1.000	1.595	1.595	1.595

Notes: The table reports country-specific Step 3 results, comparing M3 with M4; that is, the restricted model includes region/year fixed effects and basic controls, while the controlled model additionally includes parental domestic-violence controls. The outcome is attitudes toward domestic violence, and the treatment variable is lower secondary completion. The restricted and controlled columns report coefficient estimates, 95% confidence intervals, R^2 , and observations. The Oster approximate-bound columns report $\delta = 1$ and $\delta = -1$ bounds using $R_{\max} = 1.3\bar{R}$, capped at 1. The sign-change BP columns report exact Oster sign-change breakdown points computed with pweights using the pweight-supporting regssensitivity fork. $+\infty$ imposes no magnitude restriction; the other columns impose $|OVB| \leq p|\hat{\beta}_{med}|$ for $p = 10, 3, 2$. For Eswatini, the survey did not collect the item asking whether the respondent's father ever beat her mother, so its controlled model uses only the two available items measuring physical abuse of the respondent by her mother/stepmother and by her father/stepfather.

Coefficient Bounds under Unobservable Selection by Cinelli and Hazlett (2020)

Country	Restricted model				Controlled model				Cinelli–Hazlett sensitivity			
	Estimate	95% CI	R^2	N	Estimate	95% CI	R^2	N	$R^2_{Y \sim D X}$	RV	$RV_{\alpha=.05}$	$1 \times$ formal bound R^2_D/R^2_Y
Madagascar	-0.137	[-0.200, -0.075]	0.089	2,884	-0.138	[-0.200, -0.077]	0.097	2,884	1.08%	9.92%	4.49%	0.25% / 0.82%
Comoros	-0.058	[-0.125, 0.008]	0.032	832	-0.058	[-0.124, 0.008]	0.038	832	0.45%	6.49%	0.00%	0.77% / 0.62%
Benin	-0.045	[-0.080, -0.011]	0.139	3,412	-0.041	[-0.076, -0.007]	0.146	3,412	0.18%	4.17%	0.64%	0.22% / 0.91%
Honduras	-0.023	[-0.037, -0.010]	0.034	8,934	-0.023	[-0.036, -0.010]	0.034	8,934	0.17%	4.06%	1.52%	0.13% / 0.06%
Eswatini	-0.017	[-0.055, 0.021]	0.023	943	-0.019	[-0.057, 0.019]	0.030	943	0.14%	3.72%	0.00%	0.24% / 0.77%
Tonga	0.050	[-0.084, 0.183]	0.078	1,202	0.048	[-0.085, 0.182]	0.083	1,202	0.06%	2.49%	0.00%	0.43% / 0.50%
Samoa	0.169	[0.029, 0.310]	0.049	1,419	0.166	[0.024, 0.307]	0.052	1,419	0.35%	5.77%	0.90%	0.30% / 0.23%

Notes: The table reports country-specific Step 2 sensitivity statistics. Countries are ordered by the restricted-model estimate in ascending order, so more negative coefficients appear first. The restricted model includes region/year fixed effects and basic controls. The controlled model additionally includes indicators for physical abuse by the respondent's mother/stepmother and father/stepfather and whether her father ever beat her mother. Eswatini did not collect the last item, so its controlled model uses only the two parental/stepparental physical-abuse indicators. $R^2_{Y \sim D|X}$ is the weighted partial R^2 of lower-secondary completion with the domestic-violence-attitude outcome. RV is the minimum equal partial- R^2 strength that an omitted confounder would need to have with both treatment and outcome to reduce the coefficient to zero. $RV_{\alpha=.05}$ uses the robust t -statistic reported by Stata and gives the strength required to make the coefficient insignificant at the 5 percent level. The benchmark column reports the Cinelli–Hazlett formal bound for a confounder as strong as the added household-background block; raw observed-block partial R^2 values are used only as diagnostics in the plots.

Female: Lower Secondary Education Rate on Education Gradient of IPV

	Estimated education gradient			
	Baseline	+ Economic controls	+ Religion	+ Demographics
Male secondary+ share	-0.0026*** (0.0008)	-0.0026** (0.0010)	-0.0016* (0.0010)	-0.0018** (0.0009)
Female secondary+ share	0.0033*** (0.0008)	0.0031*** (0.0009)	0.0021** (0.0009)	0.0022** (0.0008)
Log GDP per capita		0.0074 (0.0078)	0.0071 (0.0077)	-0.0090 (0.0086)
Male employment rate		-0.0003 (0.0007)	0.0000 (0.0007)	0.0001 (0.0006)
Female employment rate		0.0009* (0.0005)	0.0005 (0.0005)	0.0005 (0.0004)
Predominantly Muslim			-0.0329*** (0.0121)	-0.0326*** (0.0118)
Other religion			-0.0115 (0.0192)	-0.0157 (0.0189)
Urban population (%)				0.0003 (0.0002)
Population ratio: $\leq 30/40$ s				-0.0277*** (0.0093)
Countries	73	73	73	73
Mean	-0.0517	-0.0517	-0.0517	-0.0517
Adjusted R^2	0.374	0.422	0.472	0.504

Notes: Country is the unit of analysis. The dependent variable is the first-stage country-specific coefficient for women (the estimated education gradient). Regressions use analytic weights equal to the inverse squared first-stage standard error, $1/\widehat{SE}(c)^2$. Heteroskedasticity-robust standard errors are in parentheses. Predominantly Christian countries form the omitted religion category.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Robustness Check

Male education ratio shows consistent signs across samples, with at least 10 % significance in followings.

- Two-IPV Threshold : Significance exists
- Number of IPV Justifications : Significance exists
- Using primary education as criterion
- Using upper secondary education as criterion
- Round 3, 4
- Round 5, 6: Significance exists
- Year 2005~2014 Significance exists
- Year 2015~2023
- Family Patriarchy Controls : Significance exists
- Controlling for Mean Values : Significance exists
- Exclude Extreme Cases of Education Ratio : Significance exists

Female: Education Attainment Rate by Dummies by Gender

	Estimated education gradient			
	Baseline	+ Economic controls	+ Religion	+ Demographics
Male lower-sec.: middle	-0.0459** (0.0227)	-0.0471** (0.0216)	-0.0377 (0.0227)	-0.0487** (0.0219)
Male lower-sec.: high	-0.0883** (0.0348)	-0.0824** (0.0356)	-0.0752** (0.0333)	-0.0797** (0.0316)
Female lower-sec.: middle	0.0883*** (0.0215)	0.0602*** (0.0220)	0.0462** (0.0225)	0.0528** (0.0204)
Female lower-sec.: high	0.1359*** (0.0319)	0.1010*** (0.0376)	0.0960*** (0.0344)	0.0926*** (0.0327)
Log GDP per capita		0.0233** (0.0108)	0.0160** (0.0080)	0.0000 (0.0084)
Male employment rate		-0.0002 (0.0006)	0.0002 (0.0005)	0.0003 (0.0004)
Female employment rate		0.0009* (0.0005)	0.0003 (0.0003)	0.0003 (0.0003)
Predominantly Muslim			-0.0437*** (0.0103)	-0.0416*** (0.0101)
Other religion			-0.0144 (0.0140)	-0.0155 (0.0142)
Urban population (%)				0.0005* (0.0002)
Population ratio: $\leq 30/40$ s				-0.0236** (0.0108)
Countries	73	73	73	73
Mean	-0.0517	-0.0517	-0.0517	-0.0517
Adjusted R^2	0.270	0.377	0.497	0.527

Notes: Country is the unit of analysis. The dependent variable is the first-stage country-specific coefficient for women (the estimated education gradient). The lower-secondary education shares are grouped as low (below 30, omitted), middle (30 to below 60), and high (60 or above). Regressions use analytic weights equal to the inverse squared first-stage standard error, $1/\widehat{SE}(c)^2$. Heteroskedasticity-robust standard errors are in parentheses. Predominantly Christian countries form the omitted religion category. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Female: Education Attainment by Patriarchial Society

	Low Family Patriarchy ($\leq$ mean)				High Family Patriarchy ($>$ mean)			
	Baseline	+ Economic controls	+ Religion	+ Demographics	Baseline	+ Economic controls	+ Religion	+ Demographics
Male secondary+ share	-0.0031*** (0.0009)	-0.0027* (0.0014)	-0.0027* (0.0015)	-0.0027* (0.0013)	-0.0002 (0.0010)	-0.0011 (0.0010)	-0.0008 (0.0012)	-0.0009 (0.0009)
Female secondary+ share	0.0031*** (0.0008)	0.0027** (0.0012)	0.0027** (0.0013)	0.0026** (0.0011)	0.0019** (0.0009)	0.0027** (0.0010)	0.0022 (0.0013)	0.0024** (0.0011)
Log GDP per capita		0.0048 (0.0106)	0.0036 (0.0107)	-0.0214* (0.0109)		-0.0011 (0.0103)	-0.0027 (0.0099)	-0.0148 (0.0125)
Male employment rate		0.0000 (0.0008)	0.0000 (0.0009)	0.0002 (0.0008)		-0.0008 (0.0006)	-0.0005 (0.0008)	-0.0007 (0.0008)
Female employment rate		-0.0006 (0.0007)	-0.0007 (0.0007)	-0.0007 (0.0005)		0.0010** (0.0004)	0.0007 (0.0006)	0.0008 (0.0006)
Predominantly Muslim			-0.0090 (0.0243)	-0.0016 (0.0254)			-0.0156 (0.0143)	-0.0123 (0.0150)
Other religion			0.0036 (0.0180)	0.0088 (0.0178)			-0.0204 (0.0217)	-0.0288 (0.0201)
Urban population (%)				0.0007 (0.0004)				-0.0001 (0.0004)
Population ratio: $\leq$ 30/40s				-0.0239** (0.0100)				-0.0487** (0.0171)
Countries	40	40	40	40	30	30	30	30
Mean	-0.0333	-0.0333	-0.0333	-0.0333	-0.0845	-0.0845	-0.0845	-0.0845
Adjusted R^2	0.310	0.280	0.240	0.300	0.332	0.405	0.374	0.428

Notes: Country is the unit of analysis. The dependent variable is the first-stage country-specific coefficient for women (the estimated education gradient). The Family Patriarchy split uses the unweighted country mean, 45.6543, among the 70 female main-sample countries with an observed 2019-based score; low includes equality. Columns 1–4 report the low group and columns 5–8 report the high group. Within each group, the four columns reproduce the Baseline, Economic controls, Religion, and Demographics specifications; the continuous Family Patriarchy score is not included as a regressor. Regressions use analytic weights equal to the inverse squared first-stage standard error, $1/(SE(\alpha))^2$. Heteroskedasticity robust standard errors are in parentheses. Predominantly Christian countries

Female: Married and Unmarried Samples

	Married women Estimated education gradient				Unmarried women Estimated education gradient			
	Baseline	+ Economic controls	+ Religion	+ Demographics	Baseline	+ Economic controls	+ Religion	+ Demographics
Male secondary+ share	-0.0025*** (0.0008)	-0.0024** (0.0010)	-0.0015 (0.0010)	-0.0017* (0.0009)	-0.0029*** (0.0008)	-0.0031*** (0.0009)	-0.0022** (0.0010)	-0.0025*** (0.0009)
Female secondary+ share	0.0032*** (0.0008)	0.0030*** (0.0009)	0.0021** (0.0009)	0.0021** (0.0008)	0.0037*** (0.0007)	0.0040*** (0.0009)	0.0029*** (0.0009)	0.0028*** (0.0009)
Log GDP per capita		0.0065 (0.0074)	0.0054 (0.0076)	-0.0115 (0.0094)		-0.0031 (0.0103)	-0.0034 (0.0098)	-0.0152* (0.0084)
Male employment rate		-0.0003 (0.0008)	0.0001 (0.0007)	0.0002 (0.0007)		0.0001 (0.0007)	0.0003 (0.0006)	0.0001 (0.0005)
Female employment rate		0.0009* (0.0005)	0.0005 (0.0005)	0.0005 (0.0005)		0.0007 (0.0006)	0.0000 (0.0006)	0.0002 (0.0006)
Predominantly Muslim			-0.0324** (0.0130)	-0.0316** (0.0123)			-0.0410*** (0.0147)	-0.0405** (0.0155)
Other religion			-0.0131 (0.0207)	-0.0161 (0.0210)			-0.0043 (0.0196)	-0.0140 (0.0157)
Urban population (%)				0.0004 (0.0003)				0.0003 (0.0002)
Population ratio: $\leq 30/40$ s				-0.0272** (0.0110)				-0.0358*** (0.0090)
Countries	74	74	74	74	73	73	73	73
Mean	-0.0496	-0.0496	-0.0496	-0.0496	-0.0462	-0.0462	-0.0462	-0.0462
Adjusted R^2	0.345	0.394	0.443	0.474	0.403	0.418	0.463	0.523

Notes: Country is the unit of analysis. For women, the dependent variable is the first-stage country-specific coefficient (the estimated education gradient). Columns 1–4 report the married sample and columns 5–8 report the unmarried sample. Regressions use analytic weights equal to the inverse squared first-stage standard error, $1/\widehat{SE}(c)^2$. Heteroskedasticity-robust standard errors are in parentheses. Predominantly Christian countries form the omitted religion category. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Male: Lower Secondary Education Rate on Education Gradient of IPV

	Estimated education gradient			
	Baseline	+ Economic controls	+ Religion	+ Demographics
Male secondary+ share	-0.0015** (0.0007)	-0.0014 (0.0009)	-0.0011 (0.0012)	-0.0003 (0.0010)
Female secondary+ share	0.0015** (0.0006)	0.0016 (0.0010)	0.0011 (0.0012)	0.0004 (0.0009)
Log GDP per capita		-0.0010 (0.0094)	0.0037 (0.0114)	0.0219** (0.0105)
Male employment rate		0.0004 (0.0005)	0.0004 (0.0005)	0.0007 (0.0006)
Female employment rate		0.0002 (0.0006)	0.0003 (0.0008)	-0.0001 (0.0007)
Predominantly Muslim			-0.0172 (0.0146)	-0.0106 (0.0148)
Other religion			-0.0103 (0.0222)	-0.0082 (0.0179)
Urban population (%)				-0.0011*** (0.0004)
Population ratio: $\leq 30/40$ s				0.0138 (0.0094)
Countries	42	42	42	42
Mean	-0.0305	-0.0305	-0.0305	-0.0305
Adjusted R^2	0.052	0.042	0.030	0.247

Notes: Country is the unit of analysis. The dependent variable is the first-stage country-specific coefficient for men (the estimated education gradient). Regressions use analytic weights equal to the inverse squared first-stage standard error, $1/\widehat{SE}(c)^2$. Heteroskedasticity-robust standard errors are in parentheses. Predominantly Christian countries form the omitted religion category.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Questions?

THANK YOU FOR LISTENING!!

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Female: Lower Secondary+ Education Effect on Two-IPV Threshold

	Estimated education gradient: two-IPV threshold			
	Baseline	+ Economic controls	+ Religion	+ Demographics
Male secondary+ share	-0.0027*** (0.0006)	-0.0027*** (0.0009)	-0.0015* (0.0008)	-0.0017** (0.0008)
Female secondary+ share	0.0032*** (0.0006)	0.0029*** (0.0008)	0.0018** (0.0007)	0.0018*** (0.0007)
Log GDP per capita		0.0083 (0.0075)	0.0070 (0.0065)	-0.0041 (0.0077)
Male employment rate		-0.0004 (0.0007)	-0.0001 (0.0005)	-0.0001 (0.0005)
Female employment rate		0.0010* (0.0005)	0.0004 (0.0005)	0.0004 (0.0005)
Predominantly Muslim			-0.0420*** (0.0116)	-0.0435*** (0.0117)
Other religion			-0.0091 (0.0155)	-0.0171 (0.0147)
Urban population (%)				0.0001 (0.0002)
Population ratio: $\leq 30/40$ s				-0.0266** (0.0105)
Countries	73	73	73	73
Mean	-0.0258	-0.0258	-0.0258	-0.0258
Adjusted R^2	0.399	0.452	0.543	0.564

Notes: Country is the unit of analysis. For women, the dependent variable is the country-specific gradient for the two-IPV threshold outcome. Regressions use analytic weights equal to the inverse squared first-stage standard error, $1/\widehat{SE}(c)^2$. Heteroskedasticity-robust standard errors are in parentheses. Predominantly Christian countries form the omitted religion category.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Female: Lower Secondary+ Education Effect on Justified IPV Number

	Estimated education gradient: IPV-justification count			
	Baseline	+ Economic controls	+ Religion	+ Demographics
Male secondary+ share	-0.0098*** (0.0019)	-0.0095*** (0.0031)	-0.0060** (0.0030)	-0.0066** (0.0029)
Female secondary+ share	0.0117*** (0.0019)	0.0108*** (0.0027)	0.0075*** (0.0026)	0.0076*** (0.0026)
Log GDP per capita		0.0274 (0.0247)	0.0230 (0.0228)	-0.0126 (0.0253)
Male employment rate		-0.0015 (0.0024)	-0.0003 (0.0020)	-0.0002 (0.0019)
Female employment rate		0.0032* (0.0019)	0.0014 (0.0016)	0.0013 (0.0016)
Predominantly Muslim			-0.1267*** (0.0399)	-0.1299*** (0.0407)
Other religion			-0.0227 (0.0504)	-0.0449 (0.0477)
Urban population (%)				0.0003 (0.0007)
Population ratio: $\leq 30/40$ s				-0.0808** (0.0339)
Countries	73	73	73	73
Mean	-0.0966	-0.0966	-0.0966	-0.0966
Adjusted R^2	0.437	0.484	0.550	0.566

Notes: Country is the unit of analysis. For women, the dependent variable is the country-specific gradient for the number of IPV justifications. Regressions use analytic weights equal to the inverse squared first-stage standard error, $1/\text{SE}(c)^2$. Heteroskedasticity-robust standard errors are in parentheses. Predominantly Christian countries form the omitted religion category.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Using primary education completion ratio (Female)

	Estimated primary-education gradient			
	Baseline	+ Economic controls	+ Religion	+ Demographics
Male primary-completion share	-0.0009 (0.0013)	-0.0015 (0.0011)	-0.0009 (0.0009)	-0.0005 (0.0010)
Female primary-completion share	0.0013 (0.0011)	0.0019** (0.0009)	0.0008 (0.0008)	0.0003 (0.0009)
Log GDP per capita		-0.0041 (0.0110)	0.0026 (0.0110)	0.0075 (0.0136)
Male employment rate		0.0004 (0.0007)	0.0009 (0.0006)	0.0008 (0.0006)
Female employment rate		0.0011** (0.0004)	0.0005 (0.0005)	0.0005 (0.0005)
Predominantly Muslim			-0.0502*** (0.0146)	-0.0556*** (0.0172)
Other religion			-0.0195 (0.0192)	-0.0389* (0.0207)
Urban population (%)				-0.0007* (0.0003)
Population ratio: $\leq 30/40$ s				-0.0295* (0.0148)
Countries	73	73	73	73
Mean	-0.0627	-0.0627	-0.0627	-0.0627
Adjusted R^2	0.089	0.292	0.432	0.468

Notes: Country is the unit of analysis. For women, the dependent variable is the first-stage country-specific coefficient (the estimated education gradient). Regressions use analytic weights equal to the inverse squared first-stage standard error, $1/\widehat{SE}(c)^2$. Heteroskedasticity-robust standard errors are in parentheses. Predominantly Christian countries form the omitted religion category. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Using upper secondary education completion ratio (Female)

	Estimated upper-secondary-education gradient			
	Baseline	+ Economic controls	+ Religion	+ Demographics
Male upper-secondary share	-0.0024*** (0.0008)	-0.0017** (0.0007)	-0.0005 (0.0007)	-0.0007 (0.0007)
Female upper-secondary share	0.0034*** (0.0008)	0.0022** (0.0008)	0.0011 (0.0007)	0.0013* (0.0007)
Log GDP per capita		0.0285*** (0.0081)	0.0234*** (0.0069)	0.0072 (0.0073)
Male employment rate		-0.0008 (0.0008)	-0.0001 (0.0006)	0.0001 (0.0006)
Female employment rate		0.0010 (0.0006)	0.0004 (0.0005)	0.0005 (0.0005)
Predominantly Muslim			-0.0499*** (0.0113)	-0.0452*** (0.0112)
Other religion			-0.0182* (0.0104)	-0.0260** (0.0113)
Urban population (%)				0.0001 (0.0002)
Population ratio: $\leq 30/40$ s				-0.0368*** (0.0109)
Countries	72	72	72	72
Mean	-0.0454	-0.0454	-0.0454	-0.0454
Adjusted R^2	0.352	0.458	0.588	0.626

Notes: Country is the unit of analysis. For women, the dependent variable is the first-stage country-specific coefficient (the estimated education gradient). Regressions use analytic weights equal to the inverse squared first-stage standard error, $1/\widehat{SE}(c)^2$. Heteroskedasticity-robust standard errors are in parentheses. Predominantly Christian countries form the omitted religion category. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Recent Round 3, 4 (Education Share) (Female)

	Estimated education gradient			
	Baseline	+ Economic controls	+ Religion	+ Demographics
Male secondary+ share	-0.0016 (0.0011)	-0.0019* (0.0010)	-0.0016 (0.0011)	-0.0017 (0.0010)
Female secondary+ share	0.0028** (0.0011)	0.0029*** (0.0009)	0.0024** (0.0010)	0.0023** (0.0009)
Log GDP per capita		0.0016 (0.0089)	0.0011 (0.0103)	-0.0117 (0.0093)
Male employment rate		-0.0011 (0.0010)	-0.0009 (0.0011)	-0.0006 (0.0011)
Female employment rate		0.0011** (0.0005)	0.0008 (0.0006)	0.0008 (0.0006)
Predominantly Muslim			-0.0187 (0.0173)	-0.0172 (0.0160)
Other religion			-0.0116 (0.0288)	-0.0167 (0.0272)
Urban population (%)				0.0001 (0.0003)
Population ratio: $\leq 30/40$ s				-0.0359*** (0.0085)
Countries	47	47	47	47
Mean	-0.0669	-0.0669	-0.0669	-0.0669
Adjusted R^2	0.394	0.417	0.406	0.443

Notes: Country is the unit of analysis. For women, the dependent variable is the first-stage country-specific coefficient (the estimated education gradient). Regressions use analytic weights equal to the inverse squared first-stage standard error, $1/\widehat{SE}(c)^2$. Heteroskedasticity-robust standard errors are in parentheses. Predominantly Christian countries form the omitted religion category. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Recent Round 5, 6 (Education Share) (Female)

	Estimated education gradient			
	Baseline	+ Economic controls	+ Religion	+ Demographics
Male secondary+ share	-0.0029*** (0.0008)	-0.0027*** (0.0010)	-0.0016 (0.0011)	-0.0018* (0.0010)
Female secondary+ share	0.0032*** (0.0007)	0.0026*** (0.0009)	0.0016 (0.0011)	0.0018* (0.0010)
Log GDP per capita		0.0145 (0.0090)	0.0142 (0.0089)	-0.0001 (0.0124)
Male employment rate		-0.0003 (0.0007)	0.0001 (0.0006)	0.0001 (0.0006)
Female employment rate		0.0009* (0.0005)	0.0004 (0.0005)	0.0005 (0.0005)
Predominantly Muslim			-0.0326** (0.0129)	-0.0328** (0.0129)
Other religion			-0.0061 (0.0149)	-0.0103 (0.0168)
Urban population (%)				0.0002 (0.0003)
Population ratio: $\leq 30/40$ s				-0.0222* (0.0125)
Countries	59	59	59	59
Mean	-0.0439	-0.0439	-0.0439	-0.0439
Adjusted R^2	0.348	0.419	0.479	0.488

Notes: Country is the unit of analysis. For women, the dependent variable is the first-stage country-specific coefficient (the estimated education gradient). Regressions use analytic weights equal to the inverse squared first-stage standard error, $1/\widehat{SE}(c)^2$. Heteroskedasticity-robust standard errors are in parentheses. Predominantly Christian countries form the omitted religion category. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Year 2005~2014

	Estimated education gradient			
	Baseline	+ Economic controls	+ Religion	+ Demographics
Male secondary+ share	-0.0027** (0.0011)	-0.0034*** (0.0013)	-0.0027** (0.0012)	-0.0026** (0.0010)
Female secondary+ share	0.0036*** (0.0011)	0.0045*** (0.0012)	0.0036*** (0.0011)	0.0034*** (0.0010)
Log GDP per capita		-0.0106 (0.0091)	-0.0102 (0.0102)	-0.0270** (0.0106)
Male employment rate		-0.0003 (0.0009)	-0.0000 (0.0010)	0.0001 (0.0009)
Female employment rate		0.0005 (0.0005)	0.0001 (0.0006)	0.0001 (0.0006)
Predominantly Muslim			-0.0330* (0.0186)	-0.0330* (0.0180)
Other religion			-0.0187 (0.0321)	-0.0213 (0.0305)
Urban population (%)				0.0005 (0.0004)
Population ratio: $\leq 30/40$ s				-0.0242** (0.0106)
Countries	55	55	55	55
Mean	-0.0538	-0.0538	-0.0538	-0.0538
Adjusted R^2	0.404	0.405	0.428	0.463

Notes: Country is the unit of analysis. For women, the dependent variable is the first-stage country-specific coefficient (the estimated education gradient). Regressions use analytic weights equal to the inverse squared first-stage standard error, $1/SE(c)^2$. Heteroskedasticity-robust standard errors are in parentheses. Predominantly Christian countries form the omitted religion category. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Year 2015~2023

	Estimated education gradient			
	Baseline	+ Economic controls	+ Religion	+ Demographics
Male secondary+ share	-0.0019** (0.0007)	-0.0016* (0.0009)	-0.0004 (0.0009)	-0.0005 (0.0010)
Female secondary+ share	0.0023*** (0.0007)	0.0017** (0.0008)	0.0007 (0.0010)	0.0007 (0.0010)
Log GDP per capita		0.0160 (0.0097)	0.0141 (0.0092)	0.0085 (0.0145)
Male employment rate		-0.0005 (0.0008)	-0.0001 (0.0005)	-0.0002 (0.0006)
Female employment rate		0.0011** (0.0005)	0.0005 (0.0005)	0.0005 (0.0005)
Predominantly Muslim			-0.0384*** (0.0123)	-0.0396*** (0.0129)
Other religion			-0.0072 (0.0122)	-0.0141 (0.0167)
Urban population (%)				-0.0001 (0.0004)
Population ratio: $\leq 30/40$ s				-0.0189 (0.0194)
Countries	54	54	54	54
Mean	-0.0464	-0.0464	-0.0464	-0.0464
Adjusted R^2	0.221	0.362	0.478	0.467

Notes: Country is the unit of analysis. For women, the dependent variable is the first-stage country-specific coefficient (the estimated education gradient). Regressions use analytic weights equal to the inverse squared first-stage standard error, $1/SE(c)^2$. Heteroskedasticity-robust standard errors are in parentheses. Predominantly Christian countries form the omitted religion category. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Female: Education Attainment by Gender (Overall Samples)

	Estimated education gradient				
	Baseline	+ Economic controls	+ Religion	+ Demographics	+ Family Patriarchy
Male secondary+ share	-0.0026*** (0.0008)	-0.0026** (0.0010)	-0.0016* (0.0010)	-0.0018** (0.0009)	-0.0018* (0.0009)
Female secondary+ share	0.0033*** (0.0008)	0.0031*** (0.0009)	0.0021** (0.0009)	0.0022** (0.0008)	0.0020** (0.0008)
Log GDP per capita		0.0074 (0.0078)	0.0071 (0.0077)	-0.0090 (0.0086)	-0.0096 (0.0085)
Male employment rate		-0.0003 (0.0007)	0.0000 (0.0007)	0.0001 (0.0006)	0.0001 (0.0006)
Female employment rate		0.0009* (0.0005)	0.0005 (0.0005)	0.0005 (0.0004)	0.0003 (0.0004)
Predominantly Muslim			-0.0329*** (0.0121)	-0.0326*** (0.0118)	-0.0217* (0.0123)
Other religion			-0.0115 (0.0192)	-0.0157 (0.0189)	-0.0160 (0.0190)
Urban population (%)				0.0003 (0.0002)	0.0003 (0.0003)
Population ratio: $\leq 30/40$ s				-0.0277*** (0.0093)	-0.0274*** (0.0092)
Family Patriarchy					-0.0004* (0.0003)
Countries	73	73	73	73	70
Mean	-0.0517	-0.0517	-0.0517	-0.0517	-0.0517
Adjusted R^2	0.374	0.422	0.472	0.504	0.512

Notes: Country is the unit of analysis. The dependent variable is the first-stage country-specific coefficient for women (the estimated education gradient). Columns 1–4 retain the original samples and specifications. Column 5 adds the 2019-based SIGI discrimination-in-the-family score on its original 0–100 scale, using the 2023 value only when 2019 is missing. Countries missing both values are excluded only from column 5. Regressions use analytic weights equal to the inverse squared first-stage standard error, $1/\widehat{SE}(c)^2$. Heteroskedasticity-robust standard errors are in parentheses. Predominantly Christian countries form the omitted religion category. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Education Effect Controlling for Mean Values (Female)

	Estimated education gradient			
	Baseline + mean outcome	+ Economic controls	+ Religion	+ Demographics
Male secondary+ share	-0.0009 (0.0008)	-0.0014* (0.0008)	-0.0012 (0.0009)	-0.0014* (0.0008)
Female secondary+ share	0.0011 (0.0007)	0.0015** (0.0007)	0.0014* (0.0008)	0.0015** (0.0008)
Mean IPV-justification outcome	-0.1412*** (0.0302)	-0.1388*** (0.0306)	-0.1244*** (0.0366)	-0.1149*** (0.0353)
Log GDP per capita		-0.0031 (0.0067)	-0.0023 (0.0062)	-0.0153* (0.0078)
Male employment rate		-0.0003 (0.0005)	-0.0002 (0.0006)	-0.0002 (0.0005)
Female employment rate		0.0008** (0.0003)	0.0006* (0.0003)	0.0007* (0.0003)
Predominantly Muslim			-0.0106 (0.0128)	-0.0111 (0.0123)
Other religion			-0.0028 (0.0158)	-0.0044 (0.0162)
Urban population (%)				0.0004* (0.0002)
Population ratio: $\leq 30/40$ s				-0.0191** (0.0088)
Countries	73	73	73	73
Mean	-0.0517	-0.0517	-0.0517	-0.0517
Adjusted R^2	0.534	0.568	0.560	0.577

Notes: Country is the unit of analysis. For women, the dependent variable is the first-stage country-specific coefficient (the estimated education gradient). Regressions use analytic weights equal to inverse first-stage variance. Heteroskedasticity-robust standard errors are in parentheses. Predominantly Christian countries form the omitted religion category. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Excluding Highly Skewed Cases ($EducationShare > 0.95$)

Country ID	Country	N	Share
23	Cuba	9,574	0.952
7	Barbados	1,149	0.964
2	Albania	3,400	0.980
40	Kazakhstan	31,149	0.985
88	Uzbekistan	3,469	0.992
44	Kyrgyzstan	7,300	0.994

Excluding Highly Skewed Cases ($EducationShare > 0.95$)

Country ID	Country	sex==1				sex==2			
		N	Share	Lo05	Hi95	N	Share	Lo05	Hi95
2	Albania					3,400	0.980	0	1
7	Barbados					1,149	0.964	0	1
23	Cuba	2,871	0.941	0	0	6,703	0.956	0	1
40	Kazakhstan	2,055	0.962	0	1	29,094	0.986	0	1
44	Kyrgyzstan					7,300	0.994	0	1
66	Samoa	692	0.893	0	0	2,539	0.962	0	1
88	Uzbekistan					3,469	0.992	0	1

Excluding Highly Skewed Cases ($EducationShare > 0.95$) (Female)

	Estimated education gradient			
	Baseline	+ Economic controls	+ Religion	+ Demographics
Male secondary+ share	-0.0026*** (0.0009)	-0.0025** (0.0010)	-0.0016 (0.0010)	-0.0018* (0.0009)
Female secondary+ share	0.0033*** (0.0008)	0.0030*** (0.0009)	0.0021** (0.0009)	0.0021** (0.0008)
Log GDP per capita		0.0080 (0.0078)	0.0077 (0.0078)	-0.0096 (0.0090)
Male employment rate		-0.0003 (0.0007)	0.0000 (0.0007)	0.0001 (0.0006)
Female employment rate		0.0010* (0.0005)	0.0005 (0.0005)	0.0005 (0.0005)
Predominantly Muslim			-0.0332** (0.0126)	-0.0333*** (0.0123)
Other religion			-0.0117 (0.0192)	-0.0157 (0.0190)
Urban population (%)				0.0004 (0.0003)
Population ratio: $\leq 30/40s$				-0.0285*** (0.0096)
Countries	66	66	66	66
Mean	-0.0521	-0.0521	-0.0521	-0.0521
Adjusted R^2	0.375	0.425	0.473	0.508

Notes: Country is the unit of analysis. For women, the dependent variable is the first-stage country-specific coefficient (the estimated education gradient). Regressions use analytic weights equal to the inverse squared first-stage standard error, $1/\widehat{SE}(c)^2$. Heteroskedasticity-robust standard errors are in parentheses. Predominantly Christian countries form the omitted religion category. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Family Patriarchy Robustness Summary (Female)

- Alternative outcomes and education definitions
- Overall-sample robustness analyses
- Married-sample analyses: main and robustness
- Unmarried-sample analyses: main and robustness
- Family Patriarchy as outcome: unweighted
- Family Patriarchy as outcome: population-weighted

Family Patriarchy: Alternative Outcomes and Education Definitions

	Family Patriarchy-augmented specification				
	(1) Two-IPV threshold	(2) Number of IPV justifications	(3) Upper-secondary completion ratio	(4) Primary education criterion	(5) Upper-secondary criterion
Male secondary+ share	-0.0017** (0.0008)	-0.0067** (0.0029)			
Female secondary+ share	0.0017** (0.0007)	0.0072*** (0.0026)			
Male primary-completion share				-0.0004 (0.0010)	
Female primary-completion share				0.0001 (0.0009)	
Male upper-secondary share			-0.0015 (0.0009)		-0.0008 (0.0006)
Female upper-secondary share			0.0019** (0.0009)		0.0011* (0.0007)
Log GDP per capita	-0.0038 (0.0075)	-0.0115 (0.0249)	-0.0042 (0.0075)	0.0047 (0.0135)	0.0054 (0.0073)
Male employment rate	-0.0001 (0.0005)	-0.0004 (0.0018)	0.0001 (0.0006)	0.0009 (0.0006)	0.0002 (0.0005)
Female employment rate	0.0003 (0.0004)	0.0009 (0.0016)	0.0003 (0.0004)	0.0003 (0.0005)	0.0001 (0.0005)
Predominantly Muslim	-0.0325*** (0.0119)	-0.0968** (0.0407)	-0.0211 (0.0145)	-0.0448** (0.0215)	-0.0327** (0.0132)
Other religion	-0.0171 (0.0152)	-0.0459 (0.0493)	-0.0214 (0.0179)	-0.0365* (0.0196)	-0.0244** (0.0119)
Urban population (%)	0.0001 (0.0002)	0.0002 (0.0008)	0.0001 (0.0003)	-0.0005 (0.0003)	0.0001 (0.0003)
Population ratio: $\leq 30/40$ s	-0.0264** (0.0102)	-0.0810** (0.0332)	-0.0273*** (0.0101)	-0.0281* (0.0146)	-0.0358*** (0.0111)
Family Patriarchy: 2019-based	-0.0004* (0.0002)	-0.0014* (0.0007)	-0.0004 (0.0003)	-0.0004 (0.0004)	-0.0006* (0.0003)
Countries	70	70	69	70	69
Mean	-0.0258	-0.0967	-0.0517	-0.0637	-0.0453
Adjusted R^2	0.580	0.580	0.510	0.452	0.647

Notes: Each column reports the female-sample version of the corresponding analysis and equals its full-demographics specification plus the 2019-based SIGI discrimination-in-the-family score. The 2023 score is used only when the 2019 score is unavailable. Education measures remain on separate rows because their definitions differ across columns. Each analysis retains its native estimation sample; countries missing both SIGI values are excluded from that column only. Means are recomputed using each column's weighted estimation sample. Analytic weights equal the inverse squared first-stage standard error, $1/\text{SE}(c)^2$. Heteroskedasticity-robust standard errors are in parentheses. Predominantly Christian countries form the omitted religion category. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Family Patriarchy: Overall-Sample Robustness

	Family Patriarchy-augmented specification			
	(1) Round 3, 4	(2) Round 5, 6	(3) Controlling for mean values	(4) Exclude extreme education ratios
Male secondary+ share	-0.0015 (0.0011)	-0.0018* (0.0010)	-0.0014* (0.0008)	-0.0017* (0.0009)
Female secondary+ share	0.0021* (0.0010)	0.0017* (0.0010)	0.0014* (0.0007)	0.0020** (0.0008)
Mean IPV justification value			-0.1100*** (0.0386)	
Log GDP per capita	-0.0138 (0.0104)	-0.0000 (0.0124)	-0.0151* (0.0079)	-0.0102 (0.0089)
Male employment rate	-0.0005 (0.0012)	0.0001 (0.0005)	-0.0001 (0.0005)	0.0001 (0.0006)
Female employment rate	0.0005 (0.0007)	0.0003 (0.0005)	0.0006 (0.0003)	0.0003 (0.0004)
Predominantly Muslim	-0.0153 (0.0154)	-0.0206 (0.0159)	-0.0057 (0.0131)	-0.0217 (0.0131)
Other religion	-0.0172 (0.0276)	-0.0104 (0.0172)	-0.0049 (0.0160)	-0.0159 (0.0192)
Urban population (%)	0.0002 (0.0003)	0.0002 (0.0003)	0.0004 (0.0002)	0.0004 (0.0003)
Population ratio: $\leq 30/40$ s	-0.0346*** (0.0085)	-0.0221* (0.0127)	-0.0191** (0.0089)	-0.0281*** (0.0094)
Family Patriarchy: 2019-based	-0.0003 (0.0003)	-0.0004 (0.0003)	-0.0002 (0.0002)	-0.0004* (0.0003)
Countries	46	57	70	63
Mean	-0.0673	-0.0438	-0.0517	-0.0521
Adjusted R^2	0.427	0.492	0.573	0.515

Notes: Each column reports the female-sample full-demographics specification from the indicated robustness analysis plus the 2019-based SIGI discrimination-in-the-family score; 2023 is used only as a fallback for a missing 2019 score. Native samples are retained and are not forced to a common set of countries. Countries missing both SIGI values are excluded from that column only. The mean-value column additionally controls for the country-sex mean IPV justification value, as in its source analysis. Means are recomputed on each weighted estimation sample. Analytic weights equal $1/\sqrt{\hat{E}(c)^2}$; conventional standard errors are in parentheses. Predominantly Christian countries form the omitted religion category. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Family Patriarchy: Married-Sample Robustness

	Family Patriarchy-augmented specification: married women					
	(1) Main weighted result	(2) Round 3, 4	(3) Round 5, 6	(4) Exclude extreme education ratios	(5) Controlling for mean values	(6) Upper-secondary completion ratio
Male secondary+ share	-0.0017* (0.0009)	-0.0018 (0.0012)	-0.0015 (0.0011)	-0.0016* (0.0009)	-0.0014* (0.0009)	
Female secondary+ share	0.0020** (0.0008)	0.0025** (0.0011)	0.0014 (0.0012)	0.0019** (0.0008)	0.0017** (0.0008)	
Male upper-secondary share						-0.0014 (0.0009)
Female upper-secondary share						0.0019** (0.0009)
Mean IPV justification value					-0.0824** (0.0383)	
Log GDP per capita	-0.0118 (0.0092)	-0.0209** (0.0102)	0.0048 (0.0143)	-0.0127 (0.0096)	-0.0175* (0.0094)	-0.0050 (0.0087)
Male employment rate	0.0002 (0.0007)	-0.0001 (0.0013)	0.0000 (0.0006)	0.0003 (0.0007)	0.0000 (0.0006)	0.0001 (0.0006)
Female employment rate	0.0003 (0.0005)	0.0005 (0.0007)	0.0004 (0.0006)	0.0003 (0.0005)	0.0005 (0.0004)	0.0003 (0.0005)
Predominantly Muslim	-0.0196 (0.0124)	-0.0078 (0.0154)	-0.0247 (0.0158)	-0.0202 (0.0131)	-0.0082 (0.0131)	-0.0196 (0.0148)
Other religion	-0.0167 (0.0209)	-0.0139 (0.0293)	-0.0148 (0.0188)	-0.0157 (0.0212)	-0.0077 (0.0200)	-0.0225 (0.0194)
Urban population (%)	0.0004 (0.0003)	0.0005 (0.0003)	0.0002 (0.0003)	0.0004 (0.0003)	0.0004 (0.0003)	0.0001 (0.0003)
Population ratio: ≤ 30/40s	-0.0266** (0.0107)	-0.0350*** (0.0094)	-0.0167 (0.0157)	-0.0267** (0.0110)	-0.0207* (0.0104)	-0.0260** (0.0115)
Family Patriarchy: 2019-based	-0.0005* (0.0003)	-0.0005 (0.0004)	-0.0003 (0.0003)	-0.0005* (0.0003)	-0.0004 (0.0003)	-0.0005 (0.0003)
Countries	71	47	57	64	71	69
Mean	-0.0495	-0.0579	-0.0438	-0.0502	-0.0495	-0.0496
Adjusted R ²	0.484	0.442	0.464	0.484	0.513	0.487

Notes: Each column reports the indicated married-women full-demographics specification plus the 2019-based SIGI discrimination-in-the-family score, using 2023 only as a fallback for a missing 2019 score. Native samples are retained; countries missing both values are excluded from that column only. The mean-value column retains its original mean IPV justification control, while column 6 uses the original upper-secondary completion-ratio regressors. Means are recomputed on each weighted estimation sample. Analytic weights equal $1/SE(e)^2$; conventional standard errors are in parentheses. Predominantly Christian countries form the omitted religion category. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Family Patriarchy: Unmarried-Sample Robustness

Family Patriarchy-augmented specification: unmarried women						
	(1) Main weighted result	(2) Round 3, 4	(3) Round 5, 6	(4) Exclude extreme education ratios	(5) Controlling for mean values	(6) Upper-secondary completion ratio
Male secondary+ share	-0.0024** (0.0009)	-0.0017 (0.0011)	-0.0023** (0.0010)	-0.0026*** (0.0009)	-0.0020** (0.0009)	
Female secondary+ share	0.0026*** (0.0009)	0.0020 (0.0012)	0.0026*** (0.0009)	0.0026*** (0.0009)	0.0020** (0.0009)	
Male upper-secondary share						-0.0023** (0.0010)
Female upper-secondary share						0.0025** (0.0010)
Mean IPV justification value					-0.1181** (0.0499)	
Log GDP per capita	-0.0156* (0.0085)	0.0035 (0.0138)	-0.0164 (0.0100)	-0.0075 (0.0097)	-0.0195** (0.0084)	-0.0084 (0.0073)
Male employment rate	0.0000 (0.0005)	-0.0015 (0.0010)	0.0001 (0.0005)	-0.0000 (0.0005)	-0.0001 (0.0005)	-0.0001 (0.0006)
Female employment rate	0.0001 (0.0006)	0.0009 (0.0007)	0.0001 (0.0005)	0.0002 (0.0006)	0.0002 (0.0005)	0.0001 (0.0006)
Predominantly Muslim	-0.0324** (0.0160)	-0.0437** (0.0193)	-0.0108 (0.0194)	-0.0281 (0.0178)	-0.0188 (0.0162)	-0.0279 (0.0175)
Other religion	-0.0134 (0.0163)	-0.0371 (0.0221)	0.0073 (0.0178)	-0.0126 (0.0164)	-0.0069 (0.0139)	-0.0217 (0.0142)
Urban population (%)	0.0003 (0.0003)	-0.0006 (0.0005)	0.0002 (0.0004)	0.0002 (0.0003)	0.0004 (0.0003)	-0.0000 (0.0003)
Population ratio: ≤ 30/40s	-0.0357*** (0.0097)	-0.0395*** (0.0097)	-0.0342*** (0.0126)	-0.0275*** (0.0098)	-0.0266** (0.0100)	-0.0387*** (0.0094)
Family Patriarchy: 2019-based	-0.0004 (0.0002)	0.0005 (0.0004)	-0.0005 (0.0003)	-0.0004 (0.0002)	-0.0002 (0.0002)	-0.0005 (0.0003)
Countries	70	46	57	63	70	69
Mean	-0.0462	-0.0803	-0.0343	-0.0522	-0.0462	-0.0462
Adjusted R ²	0.524	0.284	0.466	0.473	0.557	0.519

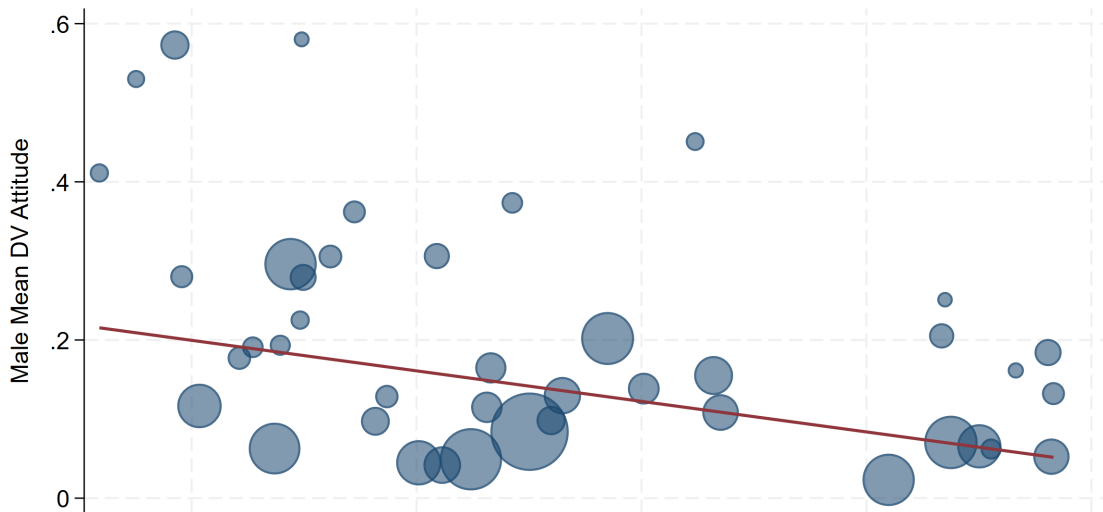
Notes: Each column reports the indicated unmarried-women full-demographics specification plus the 2019-based SIGI discrimination-in-the-family score, using 2023 only as a fallback for a missing 2019 score. Native samples are retained; countries missing both values are excluded from that column only. The mean-value column retains its original mean IPV justification control, while column 6 uses the original upper-secondary completion-ratio regressors. Means are recomputed on each weighted estimation sample. Analytic weights equal $1/\text{SE}(c)^2$; conventional standard errors are in parentheses. Predominantly Christian countries form the omitted religion category. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Family Patriarchy Outcome

	Family Patriarchy index					
	Male share only	+ Female share	+ Economic controls	+ Religion	+ Demographics	Full + Female share
Male secondary+ share	-0.4051*** (0.0656)	0.2918 (0.2630)	-0.2086*** (0.0783)	-0.2451*** (0.0846)	-0.2178** (0.0840)	0.2787 (0.2624)
Female secondary+ share		-0.6362** (0.2421)				-0.4819** (0.2403)
Log GDP per capita			-9.6073*** (2.2658)	-6.3671*** (2.3893)	-7.7300** (3.2063)	-5.6681* (3.3680)
Male employment rate			0.4354 (0.2980)	0.2887 (0.2663)	0.3004 (0.2798)	0.4092 (0.2884)
Female employment rate			-0.8007*** (0.1523)	-0.5802*** (0.1446)	-0.5603*** (0.1416)	-0.6027*** (0.1460)
Predominantly Muslim				14.3355*** (4.3327)	15.0764*** (4.4169)	14.0363*** (4.3837)
Other religion				-1.4308 (3.1997)	0.0372 (3.4277)	-1.5792 (3.7331)
Urban population (%)					0.1653 (0.1359)	0.1308 (0.1426)
Population ratio: $\leq 30/40$ s					3.0598 (4.1081)	2.9251 (4.0777)
Countries	70	70	70	70	70	70
Mean	45.6543	45.6543	45.6543	45.6543	45.6543	45.6543
Adjusted R^2	0.231	0.265	0.479	0.553	0.551	0.564

Notes: Country is the unit of analysis. The dependent variable is the 2019-based SIGI discrimination-in-the-family score (Family Patriarchy) on its original 0–100 scale; Suriname uses its 2023 value because its 2019 value is unavailable. All columns use the same 70-country female main sample. Columns 1–2 retain the original specifications; columns 3–5 omit the female secondary+ share; column 6 restores it in the full-demographics model. Regressions are unweighted. Heteroskedasticity-robust standard errors are in parentheses. Predominantly Christian countries form the omitted religion category. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Education share (male) on Men's Domestic Violence Attitude



Using Common Countries between Men and Women

	(1) Female	(2) Female	(3) Female	(4) Female	(5) Male	(6) Male	(7) Male	(8) Male
Share of Secondary+ (Male)	-0.0014* (0.0008)	-0.0003 (0.0009)	0.0004 (0.0009)	0.0004 (0.0011)	-0.0015** (0.0007)	-0.0014 (0.0009)	-0.0011 (0.0012)	-0.0003 (0.0010)
Share of Secondary+ (Female)	0.0018** (0.0007)	0.0004 (0.0009)	-0.0004 (0.0010)	-0.0004 (0.0011)	0.0015** (0.0006)	0.0016 (0.0010)	0.0011 (0.0012)	0.0004 (0.0009)
Log GDP per capita		0.0160 (0.0113)	0.0194 (0.0120)	0.0115 (0.0130)		-0.0010 (0.0094)	0.0037 (0.0114)	0.0219** (0.0105)
Employment Rate (Men)		0.0008 (0.0009)	0.0009 (0.0007)	0.0009 (0.0008)		0.0004 (0.0005)	0.0004 (0.0005)	0.0007 (0.0006)
Employment Rate (Women)		-0.0003 (0.0007)	-0.0006 (0.0006)	-0.0005 (0.0006)		0.0002 (0.0006)	0.0003 (0.0008)	-0.0001 (0.0007)
Predominant Religion: Islam			-0.0264* (0.0134)	-0.0287** (0.0134)			-0.0172 (0.0146)	-0.0106 (0.0148)
Other Religions (Non-Islam/Christian)			-0.0032 (0.0223)	-0.0112 (0.0236)			-0.0103 (0.0222)	-0.0082 (0.0179)
Urban / total pop. (%)				-0.0002 (0.0006)				-0.0011*** (0.0004)
<= 30/40s pop. ratio				-0.0218* (0.0118)				0.0138 (0.0094)
Observations	42	42	42	42	42	42	42	42
R-squared	0.133	0.212	0.276	0.313	0.098	0.159	0.196	0.412

Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Mean value of education effect is -0.0626 for female samples, and -0.0305 for male samples.

Domestic Violence population weighted

	(1) Female	(2) Female	(3) Female	(4) Female	(5) Female	(6) Male	(7) Male	(8) Male	(9) Male	(10) Male
Mean DV attitude (Male)	-0.1307** (0.0546)	-0.1230** (0.0571)	-0.1088* (0.0586)	-0.0870 (0.0607)	-0.0327 (0.0639)					
Mean DV attitude (Female)						-0.0652** (0.0261)	-0.0618** (0.0279)	-0.0669** (0.0265)	-0.0468 (0.0328)	-0.0518 (0.0308)
Share of Secondary+ (Male)		0.0001 (0.0003)		-0.0009 (0.0008)	0.0005 (0.0011)			-0.0000 (0.0003)	-0.0011 (0.0008)	-0.0004 (0.0010)
Share of Secondary+ (Female)			0.0002 (0.0003)	0.0011 (0.0007)	-0.0006 (0.0011)		0.0001 (0.0003)		0.0011 (0.0008)	0.0005 (0.0010)
Log GDP per capita					0.0119 (0.0154)					0.0175 (0.0107)
Employment Rate (Men)					0.0008 (0.0008)					0.0006 (0.0006)
Employment Rate (Women)					-0.0006 (0.0006)					0.0001 (0.0007)
Predominant Religion: Islam					-0.0271* (0.0145)					-0.0037 (0.0136)
Other Religions (Non-Islam/Christian)					-0.0076 (0.0255)					-0.0018 (0.0181)
Urban / total pop. (%)					-0.0001 (0.0006)					-0.0010** (0.0004)
<= 30/40s pop. ratio					-0.0189 (0.0128)					0.0159 (0.0098)
Observations	42	42	42	42	42	42	42	42	42	42
R-squared	0.129	0.132	0.141	0.160	0.297	0.117	0.119	0.118	0.156	0.444

Standard errors in parentheses. * p < 0.10, ** p < 0.05, *** p < 0.01.

Mean value of education effect is -0.0629 for female samples, and -0.0306 for male samples.

Population Weighted (Female)

	Estimated education gradient			
	Baseline	+ Economic controls	+ Religion	+ Demographics
Male secondary+ share	-0.0026*** (0.0009)	-0.0025** (0.0010)	-0.0016 (0.0010)	-0.0018* (0.0009)
Female secondary+ share	0.0032*** (0.0008)	0.0029*** (0.0009)	0.0020** (0.0009)	0.0021** (0.0008)
Log GDP per capita		0.0081 (0.0077)	0.0077 (0.0081)	-0.0077 (0.0087)
Male employment rate		-0.0003 (0.0007)	0.0000 (0.0007)	0.0001 (0.0006)
Female employment rate		0.0009* (0.0005)	0.0004 (0.0005)	0.0004 (0.0005)
Predominantly Muslim			-0.0326*** (0.0121)	-0.0320*** (0.0120)
Other religion			-0.0104 (0.0201)	-0.0141 (0.0199)
Urban population (%)				0.0003 (0.0002)
Population ratio: $\leq 30/40$ s				-0.0257*** (0.0095)
Countries	73	73	73	73
Mean	-0.0521	-0.0521	-0.0521	-0.0521
Adjusted R^2	0.356	0.391	0.438	0.464

Notes: Country is the unit of analysis. For women, the dependent variable is the first-stage country-specific coefficient (the estimated education gradient). First-stage country coefficients use population weights; second-stage regressions use analytic weights equal to the inverse squared first-stage standard error, $1/\widehat{SE}(c)^2$. Heteroskedasticity-robust standard errors are in parentheses. Predominantly Christian countries form the omitted religion category. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Percent (Education Effect Relative to Mean) Analysis (Female)

	Estimated education gradient relative to mean			
	Baseline + mean outcome	+ Economic controls	+ Religion	+ Demographics
Male secondary+ share	-0.0038 (0.0108)	-0.0063 (0.0100)	-0.0100 (0.0103)	-0.0097 (0.0108)
Female secondary+ share	0.0042 (0.0091)	0.0099 (0.0092)	0.0145 (0.0090)	0.0144 (0.0098)
Mean IPV-justification outcome	0.9663*** (0.2312)	0.8466*** (0.2063)	0.6548*** (0.1900)	0.6244*** (0.2053)
Log GDP per capita		-0.1630* (0.0942)	-0.2324** (0.0931)	-0.2203** (0.0954)
Male employment rate		0.0075 (0.0049)	0.0080 (0.0052)	0.0078 (0.0051)
Female employment rate		-0.0011 (0.0035)	-0.0029 (0.0042)	-0.0028 (0.0048)
Predominantly Muslim			0.0888 (0.1011)	0.0992 (0.1215)
Other religion			0.2987** (0.1294)	0.3198** (0.1360)
Urban population (%)				0.0002 (0.0043)
Population ratio: $\leq 30/40$ s				0.0508 (0.1171)
Countries	73	73	73	73
Mean	-0.4709	-0.4709	-0.4709	-0.4709
Adjusted R^2	0.153	0.188	0.223	0.199

Notes: Country is the unit of analysis. For women, the dependent variable is the country-specific coefficient divided by the corresponding country-sex outcome mean. Regressions use analytic weights equal to inverse first-stage variance. Heteroskedasticity-robust standard errors are in parentheses. Predominantly Christian countries form the omitted religion category. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Including DHS Results (Education Share by Gender)

Table 1: "Effects of Educational Attainment Rate by Gender on Returns to Lower Secondary+ Education (MICS, Country Level)"

	(1) Female	(2) Female	(3) Female	(4) Female	(5) Male	(6) Male	(7) Male	(8) Male
Share of Secondary+ (Male)	-0.0012 (0.0008)	-0.0006 (0.0009)	-0.0004 (0.0008)	-0.0005 (0.0007)	-0.0012 (0.0009)	-0.0005 (0.0008)	-0.0001 (0.0009)	0.0001 (0.0009)
Share of Secondary+ (Female)	0.0023*** (0.0008)	0.0015* (0.0009)	0.0012 (0.0008)	0.0012 (0.0008)	0.0018** (0.0007)	0.0006 (0.0010)	0.0000 (0.0010)	-0.0002 (0.0009)
Log GDP per capita		0.0123 (0.0091)	0.0130 (0.0087)	-0.0047 (0.0113)		0.0246* (0.0142)	0.0321** (0.0138)	0.0500*** (0.0132)
Employment Rate (Men)		0.0002 (0.0005)	0.0003 (0.0005)	0.0004 (0.0005)		-0.0006 (0.0005)	-0.0004 (0.0005)	-0.0002 (0.0005)
Employment Rate (Women)		0.0006 (0.0004)	0.0003 (0.0003)	0.0003 (0.0003)		0.0008* (0.0004)	0.0006 (0.0003)	0.0004 (0.0004)
Predominant Religion: Islam			-0.0382*** (0.0107)	-0.0373*** (0.0103)			-0.0243* (0.0135)	-0.0190 (0.0138)
Other Religions (Non-Islam/Christian)			0.0025 (0.0155)	-0.0013 (0.0170)			-0.0216 (0.0143)	-0.0150 (0.0148)
Urban / total pop. (%)				0.0005 (0.0004)				-0.0004 (0.0006)
<= 30/40s pop. ratio				-0.0285*** (0.0103)				0.0335** (0.0157)
Observations	117	117	117	117	53	53	53	53
R-squared	0.286	0.327	0.425	0.467	0.150	0.275	0.331	0.417

Standard errors in parentheses. * p < 0.10, ** p < 0.05, *** p < 0.01.

Upper Secondary Completion Ratio (Married) (Female)

	Estimated education gradient			
	Baseline	+ Economic controls	+ Religion	+ Demographics
Male upper-secondary share	-0.0023*** (0.0007)	-0.0021** (0.0009)	-0.0010 (0.0010)	-0.0012 (0.0010)
Female upper-secondary share	0.0035*** (0.0007)	0.0031*** (0.0010)	0.0019* (0.0010)	0.0019* (0.0010)
Log GDP per capita		0.0082 (0.0080)	0.0084 (0.0068)	-0.0037 (0.0089)
Male employment rate		-0.0003 (0.0007)	0.0001 (0.0007)	0.0001 (0.0007)
Female employment rate		0.0009* (0.0005)	0.0005 (0.0005)	0.0005 (0.0005)
Predominantly Muslim			-0.0331** (0.0129)	-0.0326** (0.0130)
Other religion			-0.0166 (0.0180)	-0.0217 (0.0194)
Urban population (%)				0.0002 (0.0003)
Population ratio: $\leq 30/40$ s				-0.0248** (0.0114)
Countries	72	72	72	72
Mean	-0.0497	-0.0497	-0.0497	-0.0497
Adjusted R^2	0.383	0.421	0.468	0.481

Notes: Country is the unit of analysis. For women, the dependent variable is the first-stage country-specific coefficient (the estimated education gradient). Regressions use analytic weights equal to the inverse squared first-stage standard error, $1/\widehat{SE}(c)^2$. Heteroskedasticity-robust standard errors are in parentheses. Predominantly Christian countries form the omitted religion category. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Recent Round 3, 4 (Married) (Female)

	Estimated education gradient			
	Baseline	+ Economic controls	+ Religion	+ Demographics
Male secondary+ share	-0.0020* (0.0011)	-0.0022* (0.0012)	-0.0019 (0.0013)	-0.0020* (0.0011)
Female secondary+ share	0.0032*** (0.0012)	0.0032*** (0.0010)	0.0029** (0.0011)	0.0028*** (0.0010)
Log GDP per capita		0.0007 (0.0072)	-0.0006 (0.0085)	-0.0180* (0.0097)
Male employment rate		-0.0008 (0.0011)	-0.0006 (0.0012)	-0.0003 (0.0013)
Female employment rate		0.0011** (0.0005)	0.0009 (0.0006)	0.0008 (0.0006)
Predominantly Muslim			-0.0138 (0.0186)	-0.0110 (0.0167)
Other religion			-0.0123 (0.0310)	-0.0126 (0.0290)
Urban population (%)				0.0004 (0.0003)
Population ratio: $\leq 30/40$ s				-0.0370*** (0.0094)
Countries	48	48	48	48
Mean	-0.0578	-0.0578	-0.0578	-0.0578
Adjusted R^2	0.408	0.430	0.412	0.449

Notes: Country is the unit of analysis. For women, the dependent variable is the first-stage country-specific coefficient (the estimated education gradient). Regressions use analytic weights equal to the inverse squared first-stage standard error, $1/\widehat{SE}(c)^2$. Heteroskedasticity-robust standard errors are in parentheses. Predominantly Christian countries form the omitted religion category. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Recent Round 5, 6 (Married) (Female)

	Estimated education gradient			
	Baseline	+ Economic controls	+ Religion	+ Demographics
Male secondary+ share	-0.0028*** (0.0009)	-0.0026** (0.0010)	-0.0014 (0.0012)	-0.0015 (0.0011)
Female secondary+ share	0.0031*** (0.0008)	0.0025** (0.0010)	0.0013 (0.0012)	0.0015 (0.0011)
Log GDP per capita		0.0160* (0.0095)	0.0167* (0.0096)	0.0047 (0.0142)
Male employment rate		-0.0003 (0.0008)	0.0001 (0.0006)	0.0000 (0.0006)
Female employment rate		0.0010* (0.0005)	0.0004 (0.0006)	0.0005 (0.0006)
Predominantly Muslim			-0.0366** (0.0139)	-0.0361** (0.0139)
Other religion			-0.0120 (0.0153)	-0.0145 (0.0185)
Urban population (%)				0.0002 (0.0003)
Population ratio: $\leq 30/40$ s				-0.0169 (0.0158)
Countries	59	59	59	59
Mean	-0.0439	-0.0439	-0.0439	-0.0439
Adjusted R^2	0.301	0.382	0.465	0.464

Notes: Country is the unit of analysis. For women, the dependent variable is the first-stage country-specific coefficient (the estimated education gradient). Regressions use analytic weights equal to the inverse squared first-stage standard error, $1/\widehat{SE}(c)^2$. Heteroskedasticity-robust standard errors are in parentheses. Predominantly Christian countries form the omitted religion category. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Education Effect Controlling for Mean Values (Married) (Female)

	Estimated education gradient			
	Baseline + mean outcome	+ Economic controls	+ Religion	+ Demographics
Male secondary+ share	-0.0011 (0.0008)	-0.0016* (0.0008)	-0.0013 (0.0009)	-0.0015* (0.0009)
Female secondary+ share	0.0015** (0.0007)	0.0020** (0.0008)	0.0018** (0.0008)	0.0018** (0.0008)
Mean IPV-justification outcome	-0.1186*** (0.0339)	-0.1215*** (0.0360)	-0.1000** (0.0393)	-0.0895** (0.0365)
Log GDP per capita		-0.0054 (0.0078)	-0.0041 (0.0073)	-0.0181* (0.0095)
Male employment rate		-0.0003 (0.0006)	-0.0001 (0.0006)	-0.0000 (0.0006)
Female employment rate		0.0008** (0.0004)	0.0006 (0.0004)	0.0006* (0.0004)
Predominantly Muslim			-0.0156 (0.0134)	-0.0161 (0.0123)
Other religion			-0.0049 (0.0196)	-0.0066 (0.0202)
Urban population (%)				0.0004 (0.0002)
Population ratio: $\leq 30/40$ s				-0.0208** (0.0104)
Countries	74	74	74	74
Mean	-0.0496	-0.0496	-0.0496	-0.0496
Adjusted R^2	0.456	0.498	0.494	0.513

Notes: Country is the unit of analysis. For women, the dependent variable is the first-stage country-specific coefficient (the estimated education gradient). Regressions use analytic weights equal to inverse first-stage variance. Heteroskedasticity-robust standard errors are in parentheses. Predominantly Christian countries form the omitted religion category. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Excluding Highly Skewed Cases ($EducationShare > 0.95$)

Country ID	Country	sex==1		sex==2	
		N	Share	N	Share
2	Albania			3,026	0.982
7	Barbados			505	0.967
40	Kazakhstan	1,946	0.964	30,367	0.986
44	Kyrgyzstan			14,242	0.994
66	Samoa	515	0.921	2,179	0.966
88	Uzbekistan			18,527	0.997

Excluding Highly Skewed Cases ($EducationShare > 0.95$) (Female)

	Estimated education gradient			
	Baseline	+ Economic controls	+ Religion	+ Demographics
Male secondary+ share	-0.0025*** (0.0009)	-0.0024** (0.0010)	-0.0015 (0.0010)	-0.0016* (0.0009)
Female secondary+ share	0.0032*** (0.0008)	0.0029*** (0.0009)	0.0020** (0.0009)	0.0021** (0.0009)
Log GDP per capita		0.0069 (0.0075)	0.0057 (0.0078)	-0.0125 (0.0098)
Male employment rate		-0.0003 (0.0008)	0.0001 (0.0007)	0.0002 (0.0007)
Female employment rate		0.0009* (0.0005)	0.0004 (0.0005)	0.0005 (0.0005)
Predominantly Muslim			-0.0333** (0.0134)	-0.0328** (0.0127)
Other religion			-0.0128 (0.0207)	-0.0152 (0.0212)
Urban population (%)				0.0004 (0.0003)
Population ratio: $\leq 30/40$ s				-0.0275** (0.0112)
Countries	67	67	67	67
Mean	-0.0503	-0.0503	-0.0503	-0.0503
Adjusted R^2	0.340	0.388	0.439	0.475

Notes: Country is the unit of analysis. For women, the dependent variable is the first-stage country-specific coefficient (the estimated education gradient). Regressions use analytic weights equal to the inverse squared first-stage standard error, $1/\widehat{SE}(c)^2$. Heteroskedasticity-robust standard errors are in parentheses. Predominantly Christian countries form the omitted religion category. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Percent Analysis (Female)

	Estimated education gradient relative to mean			
	Baseline + mean outcome	+ Economic controls	+ Religion	+ Demographics
Male secondary+ share	-0.0041 (0.0120)	-0.0070 (0.0110)	-0.0101 (0.0113)	-0.0099 (0.0117)
Female secondary+ share	0.0058 (0.0104)	0.0117 (0.0099)	0.0154 (0.0099)	0.0155 (0.0106)
Mean IPV-justification outcome	0.9228*** (0.2410)	0.7127*** (0.2372)	0.5241** (0.2015)	0.4897** (0.2116)
Log GDP per capita		-0.1726* (0.0930)	-0.2288** (0.0950)	-0.2162** (0.0938)
Male employment rate		0.0066 (0.0051)	0.0070 (0.0054)	0.0068 (0.0053)
Female employment rate		0.0003 (0.0038)	-0.0011 (0.0046)	-0.0010 (0.0050)
Predominantly Muslim			0.0769 (0.0997)	0.0892 (0.1163)
Other religion			0.2581* (0.1445)	0.2841* (0.1497)
Urban population (%)				0.0003 (0.0046)
Population ratio: $\leq 30/40$ s				0.0614 (0.1296)
Countries	74	74	74	74
Mean	-0.4315	-0.4315	-0.4315	-0.4315
Adjusted R^2	0.080	0.117	0.132	0.107

Notes: Country is the unit of analysis. For women, the dependent variable is the country-specific coefficient divided by the corresponding country-sex outcome mean. Regressions use analytic weights equal to inverse first-stage variance. Heteroskedasticity-robust standard errors are in parentheses. Predominantly Christian countries form the omitted religion category. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Including DHS Results (Married)

Table 2: "Effects of Educational Attainment Rate by Gender on Returns to Lower Secondary+ Education (MICS, Country Level)"

	(1) Female	(2) Female	(3) Female	(4) Female	(5) Male	(6) Male	(7) Male	(8) Male
Share of Secondary+ (Male)	-0.0021* (0.0010)	-0.0019* (0.0011)	-0.0011 (0.0010)	-0.0013 (0.0009)	-0.0017*** (0.0006)	-0.0019* (0.0010)	-0.0011 (0.0011)	-0.0008 (0.0010)
Share of Secondary+ (Female)	0.0028*** (0.0009)	0.0027** (0.0010)	0.0019* (0.0010)	0.0020** (0.0009)	0.0018*** (0.0006)	0.0021* (0.0012)	0.0011 (0.0012)	0.0008 (0.0010)
Log GDP per capita		0.0055 (0.0075)	0.0022 (0.0081)	-0.0192* (0.0115)		0.0018 (0.0102)	0.0121 (0.0111)	0.0256** (0.0111)
Employment Rate (Men)		-0.0001 (0.0007)	0.0004 (0.0007)	0.0005 (0.0007)		0.0001 (0.0011)	0.0002 (0.0009)	0.0002 (0.0010)
Employment Rate (Women)		0.0010* (0.0005)	0.0003 (0.0006)	0.0002 (0.0005)		0.0008 (0.0011)	0.0008 (0.0010)	0.0007 (0.0011)
Predominant Religion: Islam			-0.0386*** (0.0124)	-0.0382*** (0.0120)			-0.0303* (0.0175)	-0.0282 (0.0185)
Other Religions (Non-Islam/Christian)			-0.0088 (0.0224)	-0.0120 (0.0231)			-0.0219 (0.0219)	-0.0210 (0.0198)
Urban / total pop. (%)				0.0005 (0.0003)				-0.0006 (0.0005)
<= 30/40s pop. ratio				-0.0372*** (0.0118)				0.0158 (0.0098)
Observations	94	94	94	94	44	44	44	44
R-squared	0.271	0.361	0.424	0.483	0.113	0.213	0.306	0.387

Standard errors in parentheses. * p < 0.10, ** p < 0.05, *** p < 0.01.

Upper Secondary Completion Ratio Analysis (Female)

	Estimated education gradient			
	Baseline	+ Economic controls	+ Religion	+ Demographics
Male upper-secondary share	-0.0030*** (0.0006)	-0.0031*** (0.0008)	-0.0020** (0.0009)	-0.0022** (0.0010)
Female upper-secondary share	0.0041*** (0.0006)	0.0042*** (0.0010)	0.0029*** (0.0010)	0.0027*** (0.0010)
Log GDP per capita		0.0009 (0.0097)	0.0001 (0.0088)	-0.0075 (0.0074)
Male employment rate		-0.0002 (0.0007)	0.0000 (0.0006)	0.0001 (0.0006)
Female employment rate		0.0007 (0.0006)	0.0001 (0.0006)	0.0002 (0.0006)
Predominantly Muslim			-0.0404*** (0.0152)	-0.0391** (0.0156)
Other religion			-0.0080 (0.0173)	-0.0221* (0.0132)
Urban population (%)				-0.0000 (0.0002)
Population ratio: $\leq 30/40$ s				-0.0370*** (0.0085)
Countries	72	72	72	72
Mean	-0.0461	-0.0461	-0.0461	-0.0461
Adjusted R^2	0.431	0.434	0.474	0.517

Notes: Country is the unit of analysis. For women, the dependent variable is the first-stage country-specific coefficient (the estimated education gradient). Regressions use analytic weights equal to the inverse squared first-stage standard error, $1/\widehat{SE}(c)^2$. Heteroskedasticity-robust standard errors are in parentheses. Predominantly Christian countries form the omitted religion category. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Recent Round 3, 4 (Unmarried) (Female)

	Estimated education gradient			
	Baseline	+ Economic controls	+ Religion	+ Demographics
Male secondary+ share	-0.0017 (0.0011)	-0.0023* (0.0011)	-0.0017 (0.0011)	-0.0015 (0.0010)
Female secondary+ share	0.0028** (0.0010)	0.0028** (0.0011)	0.0020* (0.0012)	0.0017 (0.0011)
Log GDP per capita		0.0096 (0.0127)	0.0074 (0.0127)	-0.0005 (0.0122)
Male employment rate		-0.0023** (0.0010)	-0.0018* (0.0009)	-0.0015 (0.0009)
Female employment rate		0.0013* (0.0007)	0.0008 (0.0007)	0.0005 (0.0006)
Predominantly Muslim			-0.0411** (0.0201)	-0.0415** (0.0195)
Other religion			-0.0261 (0.0181)	-0.0359* (0.0197)
Urban population (%)				-0.0003 (0.0004)
Population ratio: $\leq 30/40$ s				-0.0370*** (0.0095)
Countries	47	47	47	47
Mean	-0.0790	-0.0790	-0.0790	-0.0790
Adjusted R^2	0.244	0.268	0.293	0.312

Notes: Country is the unit of analysis. For women, the dependent variable is the first-stage country-specific coefficient (the estimated education gradient). Regressions use analytic weights equal to the inverse squared first-stage standard error, $1/\widehat{SE}(c)^2$. Heteroskedasticity-robust standard errors are in parentheses. Predominantly Christian countries form the omitted religion category. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Recent Round 5, 6 (Unmarried) (Female)

	Estimated education gradient			
	Baseline	+ Economic controls	+ Religion	+ Demographics
Male secondary+ share	-0.0019** (0.0008)	-0.0021** (0.0009)	-0.0017* (0.0010)	-0.0022** (0.0010)
Female secondary+ share	0.0028*** (0.0006)	0.0030*** (0.0009)	0.0027*** (0.0009)	0.0027*** (0.0010)
Log GDP per capita		-0.0025 (0.0104)	-0.0060 (0.0096)	-0.0164 (0.0102)
Male employment rate		0.0002 (0.0006)	0.0004 (0.0006)	0.0002 (0.0005)
Female employment rate		0.0006 (0.0005)	-0.0000 (0.0005)	0.0002 (0.0006)
Predominantly Muslim			-0.0257 (0.0159)	-0.0257 (0.0166)
Other religion			0.0174 (0.0172)	0.0067 (0.0174)
Urban population (%)				0.0002 (0.0004)
Population ratio: $\leq 30/40$ s				-0.0339*** (0.0115)
Countries	59	59	59	59
Mean	-0.0344	-0.0344	-0.0344	-0.0344
Adjusted R^2	0.377	0.381	0.400	0.454

Notes: Country is the unit of analysis. For women, the dependent variable is the first-stage country-specific coefficient (the estimated education gradient). Regressions use analytic weights equal to the inverse squared first-stage standard error, $1/\widehat{SE}(c)^2$. Heteroskedasticity-robust standard errors are in parentheses. Predominantly Christian countries form the omitted religion category. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Education Effect Controlling for Mean Values (Unmarried) (Female)

	Estimated education gradient			
	Baseline + mean outcome	+ Economic controls	+ Religion	+ Demographics
Male secondary+ share	-0.0011 (0.0009)	-0.0018** (0.0008)	-0.0016* (0.0009)	-0.0020** (0.0009)
Female secondary+ share	0.0014* (0.0008)	0.0022*** (0.0008)	0.0020** (0.0009)	0.0021** (0.0009)
Mean IPV-justification outcome	-0.1784*** (0.0549)	-0.1811*** (0.0435)	-0.1554*** (0.0486)	-0.1248** (0.0471)
Log GDP per capita		-0.0114 (0.0084)	-0.0105 (0.0089)	-0.0193** (0.0082)
Male employment rate		-0.0001 (0.0005)	0.0000 (0.0005)	-0.0001 (0.0005)
Female employment rate		0.0004 (0.0004)	0.0002 (0.0005)	0.0003 (0.0005)
Predominantly Muslim			-0.0188 (0.0149)	-0.0224 (0.0152)
Other religion			-0.0008 (0.0155)	-0.0073 (0.0135)
Urban population (%)				0.0003 (0.0002)
Population ratio: $\leq 30/40$ s				-0.0265*** (0.0096)
Countries	73	73	73	73
Mean	-0.0462	-0.0462	-0.0462	-0.0462
Adjusted R^2	0.521	0.535	0.531	0.562

Notes: Country is the unit of analysis. For women, the dependent variable is the first-stage country-specific coefficient (the estimated education gradient). Regressions use analytic weights equal to inverse first-stage variance. Heteroskedasticity-robust standard errors are in parentheses. Predominantly Christian countries form the omitted religion category. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Excluding Highly Skewed Cases ($EducationShare > 0.95$)

Country ID	Country	sex==1		sex==2	
		N	Share	N	Share
7	Barbados			656	0.963
40	Kazakhstan			7,392	0.986
44	Kyrgyzstan			2,560	0.992
86	Ukraine	858	0.997	2,880	0.949
88	Uzbekistan			2,423	0.992

Excluding Highly Skewed Cases ($EducationShare > 0.95$) (Female)

	Estimated education gradient			
	Baseline	+ Economic controls	+ Religion	+ Demographics
Male secondary+ share	-0.0035*** (0.0008)	-0.0034*** (0.0009)	-0.0026*** (0.0010)	-0.0027*** (0.0009)
Female secondary+ share	0.0039*** (0.0007)	0.0037*** (0.0009)	0.0028*** (0.0009)	0.0028*** (0.0009)
Log GDP per capita		0.0061 (0.0085)	0.0053 (0.0084)	-0.0077 (0.0098)
Male employment rate		-0.0001 (0.0007)	0.0001 (0.0006)	0.0001 (0.0005)
Female employment rate		0.0009 (0.0006)	0.0003 (0.0006)	0.0003 (0.0006)
Predominantly Muslim			-0.0342** (0.0162)	-0.0370** (0.0170)
Other religion			-0.0049 (0.0174)	-0.0127 (0.0157)
Urban population (%)				0.0002 (0.0002)
Population ratio: $\leq 30/40$ s				-0.0277*** (0.0092)
Countries	66	66	66	66
Mean	-0.0521	-0.0521	-0.0521	-0.0521
Adjusted R^2	0.399	0.427	0.457	0.472

Notes: Country is the unit of analysis. For women, the dependent variable is the first-stage country-specific coefficient (the estimated education gradient). Regressions use analytic weights equal to the inverse squared first-stage standard error, $1/\widehat{SE}(c)^2$. Heteroskedasticity-robust standard errors are in parentheses. Predominantly Christian countries form the omitted religion category. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Percent Analysis (Female)

	Estimated education gradient relative to mean			
	Baseline + mean outcome	+ Economic controls	+ Religion	+ Demographics
Male secondary+ share	0.0069 (0.0222)	-0.0064 (0.0188)	-0.0110 (0.0197)	-0.0180 (0.0192)
Female secondary+ share	-0.0006 (0.0146)	0.0194 (0.0173)	0.0252 (0.0180)	0.0267 (0.0182)
Mean IPV-justification outcome	1.2389*** (0.4626)	0.8097 (0.6614)	0.7810 (0.6580)	1.2987** (0.6191)
Log GDP per capita		-0.4992** (0.2457)	-0.5862** (0.2535)	-0.7391** (0.2865)
Male employment rate		0.0164 (0.0113)	0.0175 (0.0118)	0.0149 (0.0109)
Female employment rate		-0.0096 (0.0074)	-0.0140 (0.0088)	-0.0113 (0.0077)
Predominantly Muslim			-0.0141 (0.2235)	-0.0729 (0.2465)
Other religion			0.4196** (0.2003)	0.3177 (0.2131)
Urban population (%)				0.0059 (0.0065)
Population ratio: $\leq 30/40$ s				-0.4449 (0.3038)
Countries	73	73	73	73
Mean	-0.5792	-0.5792	-0.5792	-0.5792
Adjusted R^2	0.025	0.119	0.118	0.166

Notes: Country is the unit of analysis. For women, the dependent variable is the country-specific coefficient divided by the corresponding country-sex outcome mean. Regressions use analytic weights equal to inverse first-stage variance. Heteroskedasticity-robust standard errors are in parentheses. Predominantly Christian countries form the omitted religion category. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Including DHS Results (Unmarried)

Table 3: "Effects of Educational Attainment Rate by Gender on Returns to Lower Secondary+ Education (MICS, Country Level)"

	(1) Female	(2) Female	(3) Female	(4) Female	(5) Male	(6) Male	(7) Male	(8) Male
Share of Secondary+ (Male)	-0.0021** (0.0009)	-0.0019** (0.0009)	-0.0010 (0.0009)	-0.0013 (0.0009)	-0.0025** (0.0010)	-0.0028** (0.0013)	-0.0026* (0.0015)	-0.0021 (0.0015)
Share of Secondary+ (Female)	0.0031*** (0.0008)	0.0030*** (0.0009)	0.0019** (0.0009)	0.0017* (0.0009)	0.0026*** (0.0009)	0.0027** (0.0013)	0.0026* (0.0015)	0.0021 (0.0014)
Log GDP per capita		0.0013 (0.0097)	-0.0007 (0.0100)	-0.0117 (0.0081)		0.0059 (0.0125)	0.0024 (0.0150)	0.0061 (0.0160)
Employment Rate (Men)		0.0005 (0.0006)	0.0008 (0.0006)	0.0007 (0.0006)		-0.0008* (0.0004)	-0.0007 (0.0005)	-0.0005 (0.0007)
Employment Rate (Women)		0.0005 (0.0005)	-0.0004 (0.0005)	-0.0003 (0.0005)		0.0005 (0.0006)	0.0003 (0.0007)	0.0002 (0.0008)
Predominant Religion: Islam			-0.0514*** (0.0144)	-0.0537*** (0.0158)			-0.0017 (0.0232)	-0.0007 (0.0228)
Other Religions (Non-Islam/Christian)			-0.0042 (0.0195)	-0.0177 (0.0158)			0.0100 (0.0179)	0.0061 (0.0186)
Urban / total pop. (%)				0.0002 (0.0003)				-0.0007 (0.0006)
<= 30/40s pop. ratio				-0.0425*** (0.0101)				-0.0082 (0.0272)
Observations	93	93	93	93	43	43	43	43
R-squared	0.344	0.382	0.457	0.520	0.135	0.157	0.162	0.189

Standard errors in parentheses. * p < 0.10, ** p < 0.05, *** p < 0.01.

Without Weight (Education Share by Gender) (Female)

	Estimated education gradient			
	Baseline	+ Economic controls	+ Religion	+ Demographics
Male secondary+ share	-0.0016 (0.0010)	-0.0018 (0.0013)	-0.0012 (0.0012)	-0.0012 (0.0011)
Female secondary+ share	0.0023** (0.0009)	0.0025* (0.0013)	0.0019 (0.0013)	0.0018 (0.0012)
Log GDP per capita		0.0025 (0.0158)	-0.0042 (0.0148)	-0.0080 (0.0141)
Male employment rate		-0.0005 (0.0007)	0.0000 (0.0007)	0.0001 (0.0007)
Female employment rate		-0.0002 (0.0005)	-0.0009 (0.0005)	-0.0009 (0.0006)
Predominantly Muslim			-0.0514*** (0.0180)	-0.0517*** (0.0188)
Other religion			-0.0180 (0.0238)	-0.0205 (0.0237)
Urban population (%)				-0.0001 (0.0004)
Population ratio: $\leq 30/40$ s				-0.0218 (0.0139)
Countries	73	73	73	73
Mean	-0.0722	-0.0722	-0.0722	-0.0722
Adjusted R^2	0.202	0.181	0.272	0.264

Notes: Country is the unit of analysis. For women, the dependent variable is the first-stage country-specific coefficient (the estimated education gradient). Regressions are unweighted. Heteroskedasticity-robust standard errors are in parentheses. Predominantly Christian countries form the omitted religion category. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Married Samples Without Weight (Education Share by Gender) (Female)

	Estimated education gradient			
	Baseline	+ Economic controls	+ Religion	+ Demographics
Male secondary+ share	-0.0016 (0.0010)	-0.0018 (0.0013)	-0.0012 (0.0012)	-0.0012 (0.0011)
Female secondary+ share	0.0026*** (0.0010)	0.0027** (0.0013)	0.0022* (0.0013)	0.0021* (0.0011)
Log GDP per capita		0.0019 (0.0127)	-0.0049 (0.0125)	-0.0094 (0.0117)
Male employment rate		-0.0004 (0.0007)	0.0001 (0.0006)	0.0002 (0.0006)
Female employment rate		0.0000 (0.0005)	-0.0006 (0.0005)	-0.0007 (0.0006)
Predominantly Muslim			-0.0480*** (0.0173)	-0.0482*** (0.0180)
Other religion			-0.0235 (0.0249)	-0.0261 (0.0246)
Urban population (%)				-0.0001 (0.0004)
Population ratio: $\leq 30/40$ s				-0.0238 (0.0149)
Countries	74	74	74	74
Mean	-0.0650	-0.0650	-0.0650	-0.0650
Adjusted R^2	0.287	0.261	0.332	0.326

Notes: Country is the unit of analysis. For women, the dependent variable is the first-stage country-specific coefficient (the estimated education gradient). Regressions are unweighted. Heteroskedasticity-robust standard errors are in parentheses. Predominantly Christian countries form the omitted religion category. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Unmarried Samples Without Weight (Education Share by Gender) (Female)

	Estimated education gradient			
	Baseline	+ Economic controls	+ Religion	+ Demographics
Male secondary+ share	-0.0018 (0.0012)	-0.0023 (0.0016)	-0.0014 (0.0016)	-0.0016 (0.0016)
Female secondary+ share	0.0019* (0.0010)	0.0020 (0.0016)	0.0013 (0.0016)	0.0013 (0.0016)
Log GDP per capita		0.0102 (0.0225)	-0.0007 (0.0203)	-0.0082 (0.0224)
Male employment rate		-0.0013 (0.0012)	-0.0005 (0.0011)	-0.0004 (0.0011)
Female employment rate		-0.0006 (0.0008)	-0.0016* (0.0009)	-0.0016* (0.0009)
Predominantly Muslim			-0.0707** (0.0284)	-0.0703** (0.0296)
Other religion			-0.0058 (0.0226)	-0.0077 (0.0233)
Urban population (%)				0.0000 (0.0005)
Population ratio: $\leq 30/40$ s				-0.0270 (0.0230)
Countries	73	73	73	73
Mean	-0.0873	-0.0873	-0.0873	-0.0873
Adjusted R^2	0.001	0.023	0.117	0.101

Notes: Country is the unit of analysis. For women, the dependent variable is the first-stage country-specific coefficient (the estimated education gradient). Regressions are unweighted. Heteroskedasticity-robust standard errors are in parentheses. Predominantly Christian coun-

1. Setup: Extended Education Environments

Woman j in country c chooses an expressed IPV-intolerant attitude:

$$A_{jc} \in [0, 1], \quad A_{jc} \uparrow \implies \text{stronger expressed opposition to IPV.}$$

Private belief

$$B_{jc}^F = B^F(e_{jc}^F), \quad B_e^F \equiv \frac{dB^F(e_{jc}^F)}{de_{jc}^F} > 0.$$

A woman's own education shifts her private belief toward stronger opposition to IPV.

Normative environment

$$S_c = S(E_c^M, E_c^F),$$

$$S_M \equiv \frac{\partial S_c}{\partial E_c^M} > 0, \quad S_F \equiv \frac{\partial S_c}{\partial E_c^F} \geq 0.$$

Male and female attainment shares proxy components of the environment governing which IPV-opposing attitudes can be expressed without sanction.

Threshold

$$A_{jc} \leq S_c: \text{no sanction}; \quad A_{jc} > S_c: \text{sanction cost.}$$

2. Female Educational Environment

Relative sensitivity

$$\rho = \rho(e_{jc}^F, E_c^F) > 0.$$

Sanction-free threshold

$$S_c = S(E_c^M, E_c^F), \quad S_F \geq 0.$$

A more educated female environment may also make stronger IPV opposition more socially supportable and therefore raise the sanction-free threshold.

3. Education- and Environment-Dependent Sensitivity

The woman faces two expression costs:

- private-belief mismatch: $A_{jc} \neq B^F(e_{jc}^F)$;
- normative sanction: $A_{jc} > S(E_c^M, E_c^F)$.

Allow own education and the female educational environment to affect the sensitivity parameters:

$$\pi_j = \pi(e_{jc}^F, E_c^F) > 0, \quad \phi_j = \phi(e_{jc}^F, E_c^F) > 0, \quad \rho(e_{jc}^F, E_c^F) \equiv \frac{\phi(e_{jc}^F, E_c^F)}{\pi(e_{jc}^F, E_c^F)} > 0.$$

Dividing the weighted problem by $\pi(e_{jc}^F, E_c^F)$ gives

$$A_{jc}^* = \arg \min_{A_{jc} \in [0,1]} \left[h\left(A_{jc} - B^F(e_{jc}^F)\right) + \rho(e_{jc}^F, E_c^F) g\left(\left[A_{jc} - S(E_c^M, E_c^F)\right]_+\right) \right].$$

Regularity conditions

$h(0) = h'(0) = 0$, $h'' > 0$; $g(0) = 0$, $g' > 0$, $g'' \geq 0$ for positive violations; and $h, g \in C^3$ in the interior binding region.

4. Two Cases

Case 1: Non-binding sanction

$$B^F(e_{jc}^F) \leq S(E_c^M, E_c^F).$$

Then

$$A_{jc}^* = B^F(e_{jc}^F),$$

so

$$\frac{\partial A_{jc}^*}{\partial e_{jc}^F} = B_e^F > 0, \quad \boxed{\frac{\partial^2 A_{jc}^*}{\partial e_{jc}^F \partial E_c^F} = 0.}$$

Because E_c^F enters only ρ and S , it has no local level or moderation effect when sanctions are non-binding.

Case 2: Binding sanction

$$B^F(e_{jc}^F) > S(E_c^M, E_c^F),$$

$$S(E_c^M, E_c^F) < A_{jc}^* < B^F(e_{jc}^F).$$

Expression is jointly shaped by

- private beliefs $B^F(e_{jc}^F)$;
- relative sanction sensitivity $\rho(e_{jc}^F, E_c^F)$;
- the threshold $S(E_c^M, E_c^F)$.

This is the main case below.

5. First-Order Condition

In the binding region, define

$$x_{jc} \equiv A_{jc}^* - B^F(e_{jc}^F) < 0, \quad z_{jc} \equiv A_{jc}^* - S(E_c^M, E_c^F) > 0.$$

The first-order condition is

$$h'(x_{jc}) + \rho(e_{jc}^F, E_c^F)g'(z_{jc}) = 0.$$

The second-order condition is

$$D_{jc} \equiv h''(x_{jc}) + \rho(e_{jc}^F, E_c^F)g''(z_{jc}) > 0.$$

Interpretation

The expressed attitude balances private-belief consistency against normative-sanction costs. The female education share can change both relative sanction sensitivity and the sanction-free threshold.

6. Female Education Effect

Differentiating the FOC with respect to own female education gives

$$\kappa_{jc} \equiv \frac{\partial A_{jc}^*}{\partial e_{jc}^F} = \frac{h''(x_{jc})B_e^F - \rho_e g'(z_{jc})}{D_{jc}} = A_B B_e^F + A_\rho \rho_e.$$

The education effect has two channels:

$$\underbrace{\frac{h''(x_{jc})B_e^F}{D_{jc}}}_{\text{private-belief channel}} + \underbrace{\frac{-\rho_e g'(z_{jc})}{D_{jc}}}_{\text{norm-sensitivity channel}}.$$

Environment-dependent norm-sensitivity channel

Here

$$\rho_e = \frac{\partial \rho(e_{jc}^F, E_c^F)}{\partial e_{jc}^F}.$$

If $\rho_e < 0$, own education lowers relative sanction sensitivity and raises expressed IPV opposition. If $\rho_{eF} > 0$, this positive contribution becomes smaller as E_c^F rises.

7. Female-Share Moderation

Define the two response derivatives

$$\mathcal{R}_{jc} \equiv \frac{\partial \kappa_{jc}}{\partial \rho} = B_e^F A_{B\rho} + \rho_e A_{\rho\rho},$$

$$\mathcal{C}_{jc} \equiv \frac{\partial \kappa_{jc}}{\partial S} = B_e^F A_{BS} + \rho_e A_{\rho S}.$$

Because E_c^F changes ρ_e directly and also shifts ρ and S , the chain rule gives

$$\frac{\partial \kappa_{jc}}{\partial E_c^F} = \underbrace{-\frac{g'(z_{jc})}{h''(x_{jc}) + \rho g''(z_{jc})} \rho_{eF}}_{\text{norm-sensitivity compression}} + \underbrace{\mathcal{R}_{jc} \rho_F}_{\text{sensitivity-level channel}} + \underbrace{\mathcal{C}_{jc} S_F}_{\text{threshold channel}} = A_\rho \rho_{eF} + \mathcal{R}_{jc} \rho_F + \mathcal{C}_{jc} S_F.$$

Here $\rho_F \equiv \partial \rho / \partial E_c^F$, $S_F \equiv \partial S / \partial E_c^F$, $\rho_{eF} \equiv \partial^2 \rho / (\partial e_{jc}^F \partial E_c^F)$ and $A_\rho = -g'(z_{jc}) / [h''(x_{jc}) + \rho g''(z_{jc})]$.

8. Closed-Form Moderation Components

For compactness, define

$$H_r \equiv h^{(r)}(x_{jc}), \quad G_r \equiv g^{(r)}(z_{jc}), \quad D_{jc} = H_2 + \rho G_2.$$

Second-order responses

$$\begin{aligned} A_{\rho\rho} &= \frac{2G_1G_2D_{jc} - G_1^2(H_3 + \rho G_3)}{D_{jc}^3}, \\ A_{BS} &= \frac{\rho(H_2^2G_3 + \rho G_2^2H_3)}{D_{jc}^3}, \\ A_{\rho S} &= \frac{H_2G_2D_{jc} + \rho G_1(H_3G_2 - H_2G_3)}{D_{jc}^3} = -A_{B\rho}. \end{aligned}$$

9. Circumstances for Sign

Notation. Assume the following signs for the first- and second-order responses:

How expressed attitudes respond

- $B_e^F > 0$: Own education strengthens private IPV opposition.
- $A_{B\rho} \leq 0$: Sensitivity mutes private-belief expression.
- $A_{\rho\rho} \geq 0$: Diminishing marginal effect of norm sensitivity.
- $A_{BS} > 0$: A higher threshold strengthens private-belief expression.
- $A_{\rho S} \geq 0$: A higher threshold softens sensitivity's negative effect.

How female education changes the environment

- $S_F \geq 0$: Female education raises the threshold.
- $\rho_F < 0$: Female education rate lowers sensitivity.
- $\rho_e < 0$: Female individual education lowers sensitivity.
- $\rho_{eF} > 0$: As female education rises, own education reduces sensitivity by less.

When $\rho_e < 0$, the condition $\rho_{eF} > 0$ means that a higher female education share makes ρ_e less negative: own education and the female educational environment are substitutes in reducing relative sanction sensitivity.

10. Female-Share Moderation: Sign Implications

*Using $A_\rho < 0$, $\rho_{eF} > 0$, $\rho_F < 0$, and $S_F \geq 0$, The response coefficients satisfy

$$\mathcal{R}_{jc} \equiv \frac{\partial \kappa_{jc}}{\partial \rho} = \underbrace{B_e^F A_{B\rho}}_{\leq 0} + \underbrace{\rho_e A_{\rho\rho}}_{\leq 0} \leq 0,$$

$$\mathcal{C}_{jc} \equiv \frac{\partial \kappa_{jc}}{\partial S} = \underbrace{B_e^F A_{BS}}_{> 0} + \underbrace{\rho_e A_{\rho S}}_{\leq 0}, \quad \text{sign}(\mathcal{C}_{jc}) \text{ is ambiguous.}$$

$$\boxed{\frac{\partial \kappa_{jc}}{\partial E_c^F} = \underbrace{A_\rho \rho_{eF}}_{< 0} + \underbrace{\mathcal{R}_{jc} \rho_F}_{\geq 0} + \underbrace{\mathcal{C}_{jc} S_F}_{\text{ambiguous}} .}$$

Overall implication

The overall sign is ambiguous. The female education share weakend the own-education effect precisely when

$$\mathcal{R}_{jc} \rho_F + \mathcal{C}_{jc} S_F < -A_\rho \rho_{eF}.$$

1. Setup: Extended Education Environments

Woman j in country c chooses an expressed IPV-intolerant attitude:

$$A_{jc} \in [0, 1], \quad A_{jc} \uparrow \implies \text{stronger expressed opposition to IPV.}$$

Private belief

$$B_{jc}^F = B^F(e_{jc}^F, E_c^F),$$

$$B_e^F \equiv \frac{\partial B^F}{\partial e_{jc}^F} > 0, \quad B_F^F \equiv \frac{\partial B^F}{\partial E_c^F} > 0,$$

$$B_{eF}^F \equiv \frac{\partial^2 B^F}{\partial e_{jc}^F \partial E_c^F} < 0.$$

Own education raises private IPV opposition. The female environment also shifts beliefs, especially among less-educated women.

Normative environment

$$S_c = S(E_c^M, \gamma E_c^F), \quad \gamma > 0,$$

$$S_M \equiv \frac{\partial S_c}{\partial E_c^M} > 0, \quad S_F \equiv \frac{\partial S_c}{\partial E_c^F} \geq 0.$$

The parameter γ rescales the female-attainment input, allowing the two attainment shares to enter the threshold with different relative importance.

Threshold

$$A_{jc} \leq S_c: \text{no sanction}; \quad A_{jc} > S_c: \text{sanction cost.}$$

2. Female Educational Environment

Private-belief environment

$$B_{jc}^F = B^F(e_{jc}^F, E_c^F),$$

$$B_F^F > 0, \quad \boxed{B_{eF}^F < 0}.$$

A higher female attainment share raises private IPV opposition, with a larger shift among less-educated women. Hence the private-belief gap associated with own education narrows.

Sanction-free threshold

$$S_c = S(E_c^M, \gamma E_c^F),$$

$$S_F = \gamma S_2(E_c^M, \gamma E_c^F) \geq 0.$$

A more educated female environment may also make stronger IPV opposition more socially supportable and therefore raise the sanction-free threshold.

Relative sensitivity

$$\rho = \rho(e_{jc}^F) > 0, \quad \rho_F = 0, \quad \rho_{eF} = 0.$$

The female attainment share does not directly change relative sanction sensitivity; it operates through private beliefs and the threshold.

3. Education- and Environment-Dependent Model

The woman faces two expression costs:

- private-belief mismatch: $A_{jc} \neq B^F(e_{jc}^F, E_c^F)$;
- normative sanction: $A_{jc} > S(E_c^M, \gamma E_c^F)$.

Allow own education to affect both sensitivity parameters:

$$\pi_j = \pi(e_{jc}^F) > 0, \quad \phi_j = \phi(e_{jc}^F) > 0, \quad \rho(e_{jc}^F) \equiv \frac{\phi(e_{jc}^F)}{\pi(e_{jc}^F)} > 0.$$

Dividing the weighted problem by $\pi(e_{jc}^F)$ gives

$$A_{jc}^* = \arg \min_{A_{jc} \in [0,1]} \left[h(A_{jc} - B^F(e_{jc}^F, E_c^F)) + \rho(e_{jc}^F) g\left([A_{jc} - S(E_c^M, \gamma E_c^F)]_+\right) \right].$$

Regularity conditions

$h(0) = h'(0) = 0$, $h'' > 0$; $g(0) = 0$, $g' > 0$, $g'' \geq 0$ for positive violations; and $h, g \in C^3$ in the interior binding region.

4. Two Cases

Case 1: Non-binding sanction

$$B^F(e_{jc}^F, E_c^F) \leq S(E_c^M, \gamma E_c^F).$$

Then

$$A_{jc}^* = B^F(e_{jc}^F, E_c^F),$$

so

$$\frac{\partial A_{jc}^*}{\partial e_{jc}^F} = B_e^F > 0, \quad \frac{\partial A_{jc}^*}{\partial E_c^F} = B_F^F > 0,$$

$$\frac{\partial^2 A_{jc}^*}{\partial e_{jc}^F \partial E_c^F} = B_{eF}^F < 0.$$

Because E_c^F enters private beliefs, it can have both a level effect and a moderation effect even when sanctions are non-binding.

Case 2: Binding sanction

$$B^F(e_{jc}^F, E_c^F) > S(E_c^M, \gamma E_c^F),$$

$$S(E_c^M, \gamma E_c^F) < A_{jc}^* < B^F(e_{jc}^F, E_c^F).$$

Expression is jointly shaped by

- private beliefs $B^F(e_{jc}^F, E_c^F)$;
- relative sanction sensitivity $\rho(e_{jc}^F)$;
- the threshold $S(E_c^M, \gamma E_c^F)$.

This is the main case below.

5. First-Order Condition

In the binding region, define

$$x_{jc} \equiv A_{jc}^* - B^F(e_{jc}^F, E_c^F) < 0, \quad z_{jc} \equiv A_{jc}^* - S(E_c^M, \gamma E_c^F) > 0.$$

The first-order condition is

$$h'(x_{jc}) + \rho(e_{jc}^F)g'(z_{jc}) = 0.$$

The second-order condition is

$$D_{jc} \equiv h''(x_{jc}) + \rho(e_{jc}^F)g''(z_{jc}) > 0.$$

Interpretation

The expressed attitude balances private-belief consistency against normative-sanction costs. The female attainment share can shift both the private-belief benchmark and the sanction-free threshold, but not relative sanction sensitivity directly.

6. Female Education Effect

Differentiating the FOC with respect to own female education gives

$$\kappa_{jc} \equiv \frac{\partial A_{jc}^*}{\partial e_{jc}^F} = \frac{h''(x_{jc})B_e^F - \rho_e g'(z_{jc})}{D_{jc}} = A_B B_e^F + A_\rho \rho_e.$$

Here

$$A_B \equiv \frac{h''(x_{jc})}{D_{jc}} > 0, \quad A_\rho \equiv -\frac{g'(z_{jc})}{D_{jc}} < 0, \quad \rho_e \equiv \frac{d\rho(e_{jc}^F)}{de_{jc}^F}.$$

The education effect has two channels:

$$\underbrace{\frac{h''(x_{jc})B_e^F}{D_{jc}}}_{\text{private-belief channel}} + \underbrace{\frac{-\rho_e g'(z_{jc})}{D_{jc}}}_{\text{norm-sensitivity channel}}.$$

Environment-dependent private-belief channel

$B_e^F = B_e^F(e_{jc}^F, E_c^F)$. If $B_{eF}^F < 0$, the private-belief contribution of own education becomes smaller as E_c^F rises. If $\rho_e < 0$, own education also lowers relative sanction sensitivity and raises expressed IPV opposition.

7. Female-Share Moderation

Define the two response derivatives

$$P_{jc} \equiv \frac{\partial \kappa_{jc}}{\partial B^F} = B_e^F A_{BB} + \rho_e A_{\rho B}, \quad C_{jc} \equiv \frac{\partial \kappa_{jc}}{\partial S} = B_e^F A_{BS} + \rho_e A_{\rho S}.$$

Because E_c^F changes B_e^F directly and also shifts the levels of B^F and S , the chain rule gives

$$\frac{\partial \kappa_{jc}}{\partial E_c^F} = \underbrace{A_B B_{eF}^F}_{\text{belief-gradient compression}} + \underbrace{P_{jc} B_F^F}_{\text{belief-level channel}} + \underbrace{C_{jc} S_F}_{\text{threshold channel}}.$$

Since $A_{BB} = -A_{BS}$ and $A_{\rho B} = -A_{\rho S}$,

$$P_{jc} = -C_{jc},$$

and therefore

$$\frac{\partial \kappa_{jc}}{\partial E_c^F} = A_B B_{eF}^F + C_{jc} (S_F - B_F^F), \quad S_F = \gamma S_2(E_c^M, \gamma E_c^F).$$

8. Closed-Form Moderation Components

For compactness, define

$$H_r \equiv h^{(r)}(x_{jc}), \quad G_r \equiv g^{(r)}(z_{jc}), \quad D_{jc} = H_2 + \rho G_2.$$

The relevant second-order responses are

$$\begin{aligned} A_{BS} &= \frac{\rho (H_2^2 G_3 + \rho G_2^2 H_3)}{D_{jc}^3}, & A_{BB} &= -A_{BS}, \\ A_{\rho S} &= \frac{H_2 G_2 D_{jc} + \rho G_1 (H_3 G_2 - H_2 G_3)}{D_{jc}^3}, & A_{\rho B} &= -A_{\rho S}. \end{aligned}$$

Hence

$$P_{jc} = B_e^F A_{BB} + \rho_e A_{\rho B} = - (B_e^F A_{BS} + \rho_e A_{\rho S}) = -C_{jc}.$$

Difference from the previous extension

Because E_c^F does not enter ρ , the female-share moderation no longer requires $A_{\rho\rho}$, ρ_F , or ρ_{eF} .

9. Female-Share Moderation: Sign Implications

Assume $A_B > 0$, $B_{eF}^F < 0$, $C_{jc} > 0$, and $S_M > 0$.

Male educational environment

$$\frac{\partial \kappa_{jc}}{\partial E_c^M} = C_{jc} S_M > 0.$$

A higher male attainment share strengthens the own-education effect on expressed IPV opposition.

Female educational environment

$$\frac{\partial \kappa_{jc}}{\partial E_c^F} = \underbrace{A_B B_{eF}^F}_{<0} + \underbrace{C_{jc}(S_F - B_F^F)}_{\text{relative threshold-belief shift}}.$$

When does female attainment flatten the gradient?

If $B_F^F \geq S_F$, then $\partial \kappa_{jc} / \partial E_c^F < 0$. If $S_F > B_F^F$, flattening still occurs when

$$C_{jc}(S_F - B_F^F) < -A_B B_{eF}^F.$$

Table 4: Descriptive Statistics by Current Marital Status: Country-Equal Means

Variable	Not Currently Married			Currently Married		
	Male Mean (SD)	Female Mean (SD)	Diff (t-stat)	Male Mean (SD)	Female Mean (SD)	Diff (t-stat)
Domestic Violence	0.212 (0.153)	0.245 (0.204)	-0.054 (-3.784)	0.200 (0.147)	0.289 (0.231)	-0.107 (-6.291)
Lower Secondary +	0.653 (0.183)	0.628 (0.240)	0.068 (4.018)	0.563 (0.232)	0.544 (0.276)	0.075 (5.210)
Age	31.768 (1.848)	34.408 (1.680)	-2.632 (-11.015)	36.918 (1.927)	35.601 (1.662)	1.516 (4.624)
Poorest	0.139 (0.076)	0.120 (0.052)	0.024 (2.418)	0.153 (0.038)	0.140 (0.044)	0.016 (3.928)
Second	0.151 (0.052)	0.148 (0.050)	0.007 (1.176)	0.183 (0.037)	0.169 (0.035)	0.014 (3.586)
Middle	0.180 (0.043)	0.178 (0.048)	0.003 (0.553)	0.196 (0.016)	0.188 (0.027)	0.007 (2.307)
Fourth	0.224 (0.045)	0.229 (0.033)	-0.010 (-1.533)	0.220 (0.026)	0.221 (0.021)	-0.005 (-1.466)
Richest	0.306 (0.123)	0.325 (0.131)	-0.024 (-1.775)	0.248 (0.051)	0.283 (0.095)	-0.032 (-5.437)
Urban	0.565 (0.164)	0.639 (0.169)	-0.041 (-2.904)	0.468 (0.166)	0.527 (0.178)	-0.013 (-1.443)

Note: Argentina does not report the marital status variable; therefore, one country is missing when splitting by current marital status.

Gender Gap in Domestic Violence

Table 5: Gender Gap in Domestic Violence

Country	Male	Fem	Diff	SE	Country	Male	Fem	Diff	SE
Mali	0.411	0.765	-0.354*	(0.016)	Malawi	0.063	0.140	-0.077*	(0.006)
Gambia	0.129	0.481	-0.353*	(0.013)	Sao Tome and Principe	0.115	0.173	-0.058*	(0.011)
Sierra Leone	0.306	0.565	-0.259*	(0.013)	Mongolia	0.070	0.126	-0.055*	(0.005)
Tonga	0.162	0.412	-0.250*	(0.023)	Fiji	0.205	0.246	-0.040*	(0.013)
Cote d'Ivoire	0.177	0.410	-0.233*	(0.014)	Suriname	0.042	0.074	-0.032*	(0.007)
Chad	0.530	0.748	-0.218*	(0.017)	Guyana	0.098	0.128	-0.030*	(0.009)
Congo DR	0.451	0.664	-0.213*	(0.015)	Nepal	0.306	0.329	-0.023*	(0.011)
Zimbabwe	0.155	0.363	-0.208*	(0.008)	Lesotho	0.193	0.209	-0.015	(0.014)
Ghana	0.139	0.307	-0.168*	(0.009)	Belize	0.045	0.059	-0.014*	(0.006)
Central Africa Republic	0.573	0.739	-0.166*	(0.011)	Eswatini	0.130	0.142	-0.012	(0.008)
Congo	0.373	0.533	-0.159*	(0.015)	Montenegro	0.052	0.063	-0.011	(0.009)
Viet Nam	0.108	0.267	-0.159*	(0.008)	Honduras	0.049	0.057	-0.008	(0.005)
Togo	0.225	0.379	-0.153*	(0.016)	Tunisia	0.165	0.165	0.000	(0.009)
Benin	0.117	0.256	-0.140*	(0.008)	Thailand	0.084	0.083	0.001	(0.004)
Samoa	0.251	0.386	-0.135*	(0.021)	Cuba	0.023	0.016	0.007	(0.005)
Madagascar	0.279	0.413	-0.134*	(0.012)	Cameroon	0.362	0.351	0.011	(0.014)
Laos	0.296	0.416	-0.121*	(0.006)	Bosnia Herzegovina	0.066	0.055	0.011	(0.007)
Nigeria	0.202	0.320	-0.118*	(0.006)	Vanuatu	0.580	0.564	0.016	(0.021)
Guinea-Bissau	0.280	0.381	-0.101*	(0.014)	Serbia	0.062	0.040	0.022	(0.015)
Mauritania	0.191	0.290	-0.099*	(0.013)	Moldova	0.132	0.110	0.023*	(0.013)
Comoros	0.097	0.181	-0.084*	(0.012)	Kazakhstan	0.184	0.132	0.053*	(0.011)

Gender Difference of Coefficients between Men and Women 2

Table 6: Significant Differences in Beta between Fathers and Mothers by Country

Country	Years (Dad)	Years (Mom)	Beta (Dad)	SE (Dad)	Obs (Dad)	Beta (Mom)	SE (Mom)	Obs (Mom)	Diff (Dad-Mom)
Nepal	2019	2014, 2019	-0.1706***	0.0211	3,624	-0.1214***	0.0121	13,375	-0.0492**
Lao People's Democratic Republic	2011, 2017	2011, 2017	-0.0112	0.0098	11,943	-0.0348***	0.0085	22,850	0.0236*
Suriname	2018	2006, 2010, 2018	-0.0153	0.0178	1,572	-0.0470***	0.0062	11,088	0.0317*
Nigeria	2016, 2021	2011, 2016, 2021	-0.0386***	0.0112	14,881	-0.0736***	0.0075	39,862	0.0350***
Tunisia	2018, 2023	2011, 2018, 2023	-0.0414**	0.0184	2,605	-0.0823***	0.0073	17,014	0.0409**
Malawi	2013, 2019	2013, 2019	-0.0079	0.0098	5,212	-0.0620***	0.0086	21,454	0.0541***
Cameroon	2014	2014	-0.1097***	0.0252	2,324	-0.1642***	0.0183	4,627	0.0545*
Comoros	2022	2022	0.0040	0.0180	1,393	-0.0525***	0.0176	3,056	0.0565**
Eswatini	2010, 2014, 2021	2010, 2014, 2021	-0.0211	0.0147	3,382	-0.0776***	0.0121	6,363	0.0566***
Guyana	2014, 2019	2006, 2014, 2019	-0.0128	0.0194	2,407	-0.0702***	0.0107	9,803	0.0574***
Gambia	2018	2005, 2010, 2018	-0.0310	0.0242	1,427	-0.1369***	0.0153	7,258	0.1059***
Togo	2010, 2017	2006, 2010, 2017	-0.0370	0.0343	1,324	-0.1433***	0.0217	5,725	0.1062***
Cote d'Ivoire	2016	2006, 2016	-0.0422*	0.0245	1,881	-0.1561***	0.0196	5,326	0.1139***
Zimbabwe	2014	2009, 2014	0.0119	0.0157	3,534	-0.1114***	0.0099	14,539	0.1233***
Guinea-Bissau	2014, 2018	2006, 2014, 2018	-0.0335	0.0284	2,525	-0.1578***	0.0200	6,194	0.1242***
Viet Nam	2020	2006, 2010, 2013, 2020	0.0016	0.0179	3,289	-0.1550***	0.0102	13,738	0.1566***

Notes: This table includes countries with statistically significant Dad-Mom differences at the 10 percent level. Diff = $\hat{\beta}_{dad} - \hat{\beta}_{mom}$. Stars indicate significance levels: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

IPV Attitude (male) on IPV Attitude (Female)

	(1) Female	(2) Female	(3) Female	(4) Female	(5) Female
Mean DV attitude (Male)	1.3846*** (0.0853)	1.2806*** (0.0966)	1.2441*** (0.0928)	1.2176*** (0.0937)	1.0745*** (0.1302)
Share of Secondary+ (Male)		-0.0013** (0.0005)		0.0023 (0.0016)	0.0003 (0.0019)
Share of Secondary+ (Female)			-0.0014*** (0.0004)	-0.0035** (0.0016)	-0.0008 (0.0016)
Log GDP per capita					-0.0400* (0.0227)
Employment Rate (Men)					-0.0014 (0.0015)
Employment Rate (Women)					0.0016 (0.0011)
Predominant Religion: Islam					0.0186 (0.0264)
Other Religions (Non-Islam/Christian)					0.0215 (0.0257)
Urban / total pop. (%)					0.0005 (0.0009)
<= 30/40s pop. ratio					0.0361 (0.0261)
Observations	42	42	42	42	42
R-squared	0.820	0.849	0.858	0.864	0.904

Dependent variable is Mean DV attitude (Female). Higher values indicate greater IPV tolerance.

Regressions use female attitude mean precision weights. Standard errors in parentheses. * p < 0.10, ** p < 0.05, *** p < 0.01.

Country-by-IPV-Item : The Sign of ρ

First stage: female country–item education coefficient

$$Y_{isck}^F = \alpha_{ck} + \tau_{ck}^F D_{isc}^F + X_{isc}' \Gamma_{ck} + \lambda_{r(i)}^c + \eta_{t(i)} + \varepsilon_{isck}^F.$$

Second stage

$$\hat{\tau}_{ck}^F = \alpha_c + \delta_k + \beta M_{ck}^M + u_{ck}$$

First-stage variables

- Y_{isck}^F : equals one if woman i justifies IPV in item k
- D_{isc}^F : lower-secondary-or-above education
- X_{isc} : urban residence, age and age squared, wealth quartiles, and current marital status
- $\lambda_{r(i)}^c$ and $\eta_{t(i)}$: region and survey-year fixed effects

Second-stage variables

- $\hat{\tau}_{ck}^F$: estimated female education coefficient in country c , item k
- M_{ck}^M : weighted male IPV-acceptance share in country–IPV item cells
- α_c and δ_k : country and IPV-item fixed effects
- w_{ck} : inverse first-stage variance weight
- Second-stage standard errors are clustered by country

3, Country-by-IPV-Item Analysis (Sign of ρ)

	(1) No FE	(2) Item FE	(3) Country FE	(4) Main FE	(5) + baseline	(6) Unweighted	(7) Capped	(8) Strict	(9) Exact survey	(10) Normalized	(11) Adjusted male
Male item-specific IPV acceptance	-0.3082*** (0.0473)	-0.2974*** (0.0498)	-0.3277*** (0.0571)	-0.3121*** (0.0753)	0.1088 (0.0999)	-0.1875* (0.1001)	-0.3048*** (0.0761)	-0.3121*** (0.0753)	-0.2857*** (0.0737)	-0.3081*** (0.0779)	
Composition-adjusted male acceptance											-0.2901*** (0.0809)
Female low-education baseline					-0.3406*** (0.0585)						
Cells	210	210	210	210	210	210	210	210	295	210	210
R-squared	0.490	0.501	0.818	0.821	0.874	0.766	0.810	0.821	0.828	0.820	0.814

tau is the female item-specific lower-secondary-or-above coefficient; male acceptance is the weighted justified share.

Inverse-variance analytic weights equal the reciprocal of $se(\tau)$ squared unless otherwise stated; standard errors are clustered by country.

The main model includes country and item fixed effects; matched male and female survey cells only.

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Country-by-IPV-Item : Macro-Control Robustness

	(1) Item FE	(2) + education	(3) + economy	(4) + religion	(5) + demography
Male item-specific IPV acceptance	-0.2974*** (0.0498)	-0.2353*** (0.0509)	-0.2173*** (0.0509)	-0.2022*** (0.0506)	-0.1907*** (0.0480)
Cells	210	210	210	210	210
R-squared	0.501	0.555	0.598	0.616	0.654

Supplemental ecological models; no country fixed effects.

All models include item fixed effects and cluster standard errors by country.

Inverse-variance analytic weights equal the reciprocal of $se(\tau_{ij})$ squared.

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Empirical Strategy: Omitted Variable Bias

Observable robustness check

$$Y_{isr}^c = \alpha + \tau^{Edu} Edu_{isr}^c + K_{isr}^c{}' \Gamma + F_{isr}^c{}' \Pi \\ + \delta_r^c + \omega_s^c + \epsilon_{isr}^c$$

Oster bound for unobservables

$$\tau_c^{Oster} \simeq \tau_c^1 - \delta(\tau_c^0 - \tau_c^1) \frac{R_{\max} - R_c^1}{R_c^1 - R_c^0}$$

Main concern

- Education may proxy for unobserved early-life family environment.
- Key omitted factors: The experience of domestic violence within the family in childhood.
- F_{isr}^c adds observable family-environment controls/proxies (Whether the respondent's experienced domestic violence from parents after age 15, and Whether the respondent's observed the fathers violence to the mother).

Implementation

- (τ_c^0, R_c^0) : baseline model; (τ_c^1, R_c^1) : model with family controls.
- Use $R_{\max} = \min\{1.3R_c^1, 1\}$ and $\delta = 1, -1$ following Oster-style proportional selection.
- Report country-specific bounds for selected countries; robustness requires preserving the sign and excluding zero.

Problem of Oster Bound (Matthew A Masten and Poirier 2022)

Conventional interpretation: explain-away breakdown point

$$\delta^{EA} = \inf \{ |\delta| : 0 \in B_I(\delta, R_{\text{long}}^2) \}.$$

- This asks: *How strong must selection on unobservables be to make the coefficient exactly zero?*
- Empirical papers often interpret a large δ^{EA} as evidence that the result is robust to omitted variables.

Key problem: sign robustness is a different question

$$\delta^{SC} = \inf \{ |\delta| : \exists b \in B_I(\delta, R_{\text{long}}^2) \text{ with the opposite sign of } \beta_{\text{med}} \}.$$

- The coefficient can change sign before it can be driven exactly to zero.
- Under Oster's Proposition 2, feasible values of β_{long} solve a cubic equation:

$$f(\beta_{\text{med}} - b, \delta, R_{\text{long}}^2) = 0.$$

- Therefore, for the same δ , multiple values of β_{long} may be consistent with the data and assumptions.

What Should We Do Instead?

Main correction within the Oster framework

- Report the sign-change BP:

$$\delta^{SC} = \inf\{|\delta| : \exists b \in B_I(\delta, R_{\text{long}}^2) \text{ with opposite sign}\}.$$

- Optionally impose transparent OVB magnitude restrictions:

$$|\beta_{\text{long}} - \beta_{\text{med}}| \leq M, \quad M = p|\hat{\beta}_{\text{med}}|, \quad p = 10, 3, 2.$$

Alternative sensitivity frameworks

- Cinelli, Ferwerda, and Hazlett (2024): replace Oster's δ with partial- R^2 sensitivity parameters.
- Diegert, Matthew A. Masten, and Poirier (2022): use r_X^2 and r_Y^2 to separately bound unobserved selection in the treatment and outcome equations.

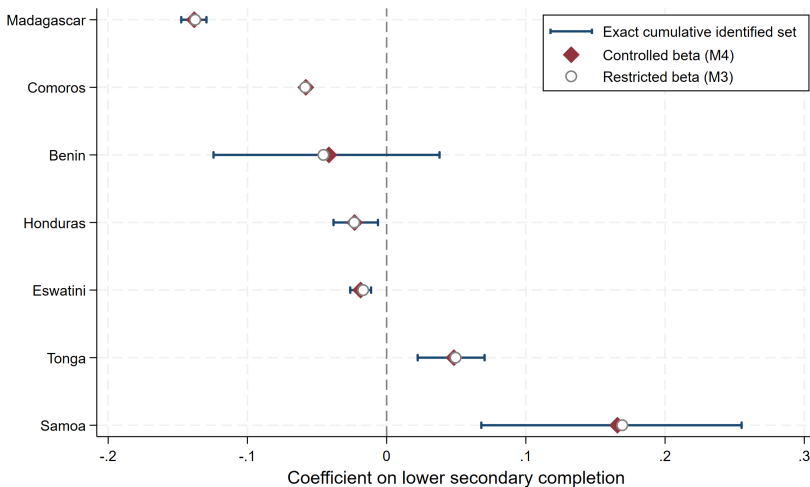
Oster Bound All Steps (Overall Samples)

Table 7: Coefficient Bounds under Unobservable Selection, by Country and Specification Step

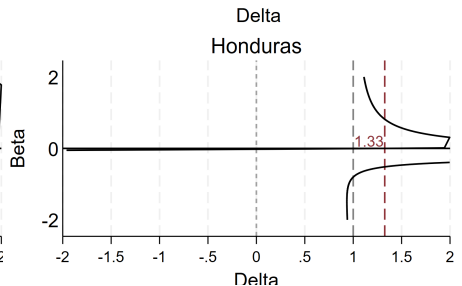
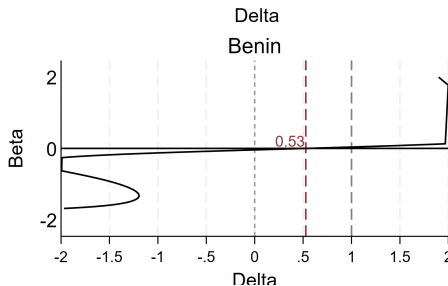
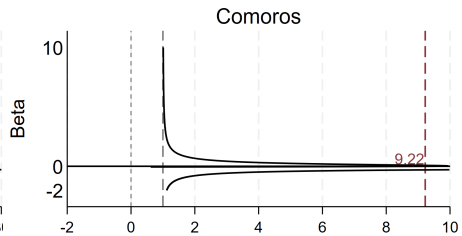
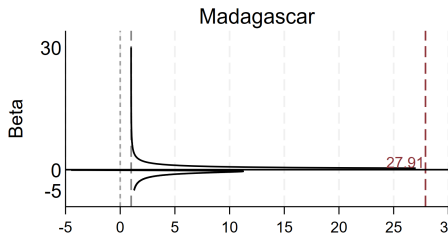
Country	Step	Restricted model				Controlled model				Oster approx. bounds		Sign-change BP			
		Estimate	95% CI	R^2	N	Estimate	95% CI	R^2	N	$\delta = 1$	$\delta = -1$	$+\infty$	$10 \hat{\beta} $	$3 \hat{\beta} $	$2 \hat{\beta} $
Madagascar	Step 1: Basic	-0.146	[-0.199, -0.094]	0.084	2,884	-0.137	[-0.200, -0.075]	0.089	2,884	-0.092	-0.183	0.229	0.229	0.229	0.229
	Step 2: Parent D.V.	-0.137	[-0.200, -0.075]	0.089	2,884	-0.138	[-0.200, -0.077]	0.097	2,884	-0.141	-0.136	0.997	4.085	27.911	27.911
	Step 3: Full	-0.146	[-0.199, -0.094]	0.084	2,884	-0.138	[-0.200, -0.077]	0.097	2,884	-0.119	-0.157	0.377	0.377	0.377	0.377
Comoros	Step 1: Basic	-0.058	[-0.118, 0.003]	0.024	832	-0.058	[-0.125, 0.008]	0.032	832	-0.059	-0.058	0.997	0.997	1.007	1.020
	Step 2: Parent D.V.	-0.058	[-0.125, 0.008]	0.032	832	-0.058	[-0.124, 0.008]	0.038	832	-0.057	-0.059	1.000	2.474	8.091	9.223
	Step 3: Full	-0.058	[-0.118, 0.003]	0.024	832	-0.058	[-0.124, 0.008]	0.038	832	-0.058	-0.058	1.000	1.040	1.278	1.412
Benin	Step 1: Basic	-0.076	[-0.109, -0.043]	0.127	3,412	-0.045	[-0.080, -0.011]	0.139	3,412	0.063	-0.154	0.118	0.118	0.118	0.118
	Step 2: Parent D.V.	-0.045	[-0.080, -0.011]	0.139	3,412	-0.041	[-0.076, -0.007]	0.146	3,412	-0.019	-0.064	0.528	0.528	0.528	0.528
	Step 3: Full	-0.076	[-0.109, -0.043]	0.127	3,412	-0.041	[-0.076, -0.007]	0.146	3,412	0.037	-0.120	0.146	0.146	0.146	0.146
Honduras	Step 1: Basic	-0.047	[-0.058, -0.036]	0.021	8,934	-0.023	[-0.037, -0.010]	0.034	8,934	-0.004	-0.042	0.484	0.484	0.484	0.484
	Step 2: Parent D.V.	-0.023	[-0.037, -0.010]	0.034	8,934	-0.023	[-0.036, -0.010]	0.034	8,934	-0.019	-0.027	1.000	1.325	1.325	1.325
	Step 3: Full	-0.047	[-0.058, -0.036]	0.021	8,934	-0.023	[-0.036, -0.010]	0.034	8,934	-0.004	-0.042	0.487	0.487	0.487	0.487
Eswatini	Step 1: Basic	-0.035	[-0.069, -0.000]	0.009	943	-0.017	[-0.055, 0.021]	0.023	943	-0.008	-0.026	1.000	1.096	1.096	1.096
	Step 2: Parent D.V.	-0.017	[-0.055, 0.021]	0.023	943	-0.019	[-0.057, 0.019]	0.030	943	-0.021	-0.016	0.630	2.369	2.369	2.369
	Step 3: Full	-0.035	[-0.069, -0.000]	0.009	943	-0.019	[-0.057, 0.019]	0.030	943	-0.012	-0.026	1.000	1.527	1.527	1.527
Tonga	Step 1: Basic	0.015	[-0.115, 0.145]	0.021	1,202	0.050	[-0.084, 0.183]	0.078	1,202	0.064	0.035	0.950	1.893	1.893	1.893
	Step 2: Parent D.V.	0.050	[-0.084, 0.183]	0.078	1,202	0.048	[-0.085, 0.182]	0.083	1,202	0.042	0.055	1.000	1.725	1.725	1.725
	Step 3: Full	0.015	[-0.115, 0.145]	0.021	1,202	0.048	[-0.085, 0.182]	0.083	1,202	0.062	0.035	0.963	2.017	2.017	2.017
Samoa	Step 1: Basic	0.160	[0.023, 0.296]	0.040	1,419	0.169	[0.029, 0.310]	0.049	1,419	0.184	0.154	0.983	2.021	4.762	4.762
	Step 2: Parent D.V.	0.169	[0.029, 0.310]	0.049	1,419	0.166	[0.024, 0.307]	0.052	1,419	0.141	0.190	1.000	1.595	1.595	1.595
	Step 3: Full	0.160	[0.023, 0.296]	0.040	1,419	0.166	[0.024, 0.307]	0.052	1,419	0.174	0.158	0.995	2.137	11.590	17.350

Notes: The table reports country-specific stepwise Oster approximate bounds. The outcome is attitudes toward domestic violence, and the treatment is lower secondary completion. Columns report estimates, 95% confidence intervals, R^2 , and observations for the restricted and controlled models. Step 1 compares the region-fixed-effects model with the model adding basic controls. Step 2 compares the basic-controls model with the model further adding parental domestic-violence controls. Step 3 compares the region-fixed-effects model with the full model. Oster approximate bounds are reported for $\delta = 1$ and $\delta = -1$, using $R_{\max} = 1.3\bar{R}$, capped at 1. The sign-change BP columns report exact Oster sign-change breakdown points computed with pweights using the pweight-supporting regnsensitivity fork. $+\infty$ imposes no magnitude restriction; the other columns impose $|OVB| \leq p|\hat{\beta}_{\text{med}}|$ for $p = 10, 3, 2$. For Eswatini, the survey did not collect the item asking whether the respondent's father ever beat her mother, so its controlled model uses only the two available items measuring physical abuse of the respondent by her mother/stepmother and by her father/stepfather.

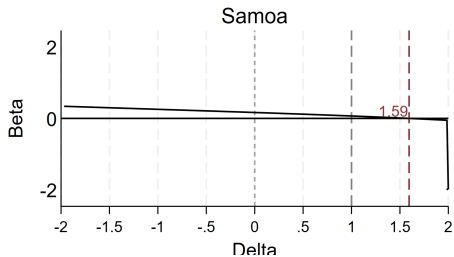
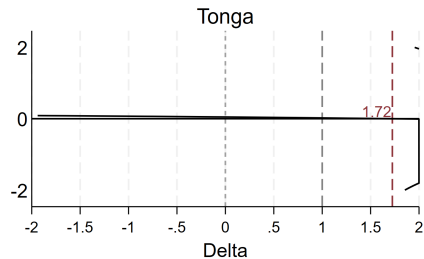
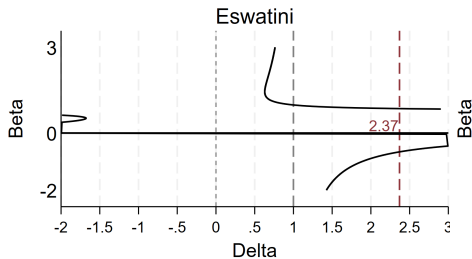
Envelope of the exact cumulative identified set



Range of All possible β_{long} 1



Range of All possible β_{long} 2



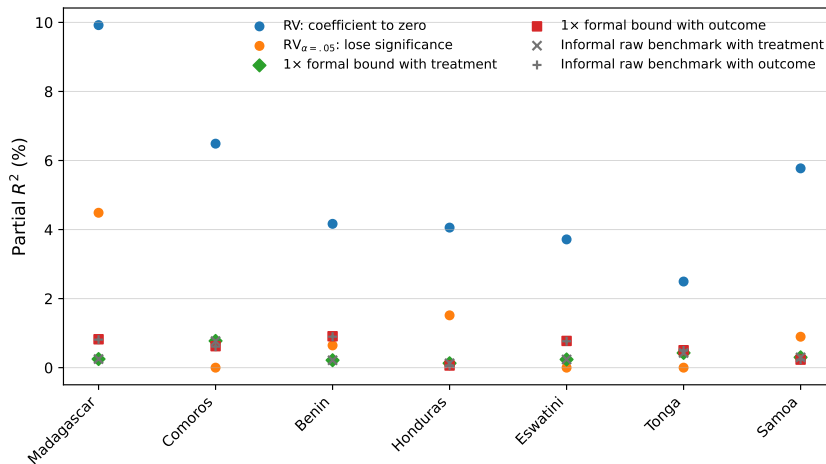
Coefficient Bounds under Cineli, Hazlett (2020)

Table 8: Cinelli–Hazlett Sensitivity Analysis by Country and Specification Step

Country	Step	Controlled estimate	95% CI	R^2	N	$R^2_{Y \sim D X}$	RV	$RV_{\alpha=.05}$	1× formal bound R^2_D/R^2_Y	Raw adj. estimate
Madagascar	FE → Basic	-0.137	[-0.200, -0.075]	0.089	2,884	1.06%	9.85%	4.31%	40.63% / 1.36%	-0.074
	Basic → Parent DV	-0.138	[-0.200, -0.077]	0.097	2,884	1.08%	9.92%	4.49%	0.25% / 0.82%	-0.132
	FE → Full	-0.138	[-0.200, -0.077]	0.097	2,884	1.08%	9.92%	4.49%	40.98% / 3.34%	-0.039
Comoros	FE → Basic	-0.058	[-0.125, 0.008]	0.032	832	0.46%	6.54%	0.00%	27.35% / 1.39%	-0.018
	Basic → Parent DV	-0.058	[-0.124, 0.008]	0.038	832	0.45%	6.49%	0.00%	0.77% / 0.62%	-0.052
	FE → Full	-0.058	[-0.124, 0.008]	0.038	832	0.45%	6.49%	0.00%	28.34% / 2.52%	-0.004
Benin	FE → Basic	-0.045	[-0.080, -0.011]	0.139	3,412	0.22%	4.54%	1.02%	16.66% / 1.92%	0.001
	Basic → Parent DV	-0.041	[-0.076, -0.007]	0.146	3,412	0.18%	4.17%	0.64%	0.22% / 0.91%	-0.037
	FE → Full	-0.041	[-0.076, -0.007]	0.146	3,412	0.18%	4.17%	0.64%	16.92% / 3.22%	0.018
Honduras	FE → Basic	-0.023	[-0.037, -0.010]	0.034	8,934	0.17%	4.10%	1.55%	41.02% / 3.09%	0.017
	Basic → Parent DV	-0.023	[-0.036, -0.010]	0.034	8,934	0.17%	4.06%	1.52%	0.13% / 0.06%	-0.023
	FE → Full	-0.023	[-0.036, -0.010]	0.034	8,934	0.17%	4.06%	1.52%	41.20% / 3.25%	0.018
Eswatini	FE → Basic	-0.017	[-0.055, 0.021]	0.023	943	0.12%	3.34%	0.00%	16.25% / 1.89%	0.006
	Basic → Parent DV	-0.019	[-0.057, 0.019]	0.030	943	0.14%	3.72%	0.00%	0.24% / 0.77%	-0.017
	FE → Full	-0.019	[-0.057, 0.019]	0.030	943	0.14%	3.72%	0.00%	16.52% / 2.99%	0.010
Tonga	FE → Basic	0.050	[-0.084, 0.183]	0.078	1,202	0.07%	2.55%	0.00%	3.16% / 6.59%	-0.033
	Basic → Parent DV	0.048	[-0.085, 0.182]	0.083	1,202	0.06%	2.49%	0.00%	0.43% / 0.50%	0.040
	FE → Full	0.048	[-0.085, 0.182]	0.083	1,202	0.06%	2.49%	0.00%	3.60% / 7.22%	-0.043
Samoa	FE → Basic	0.169	[0.029, 0.310]	0.049	1,419	0.37%	5.89%	1.06%	2.14% / 1.01%	0.129
	Basic → Parent DV	0.166	[0.024, 0.307]	0.052	1,419	0.35%	5.77%	0.90%	0.30% / 0.23%	0.158
	FE → Full	0.166	[0.024, 0.307]	0.052	1,419	0.35%	5.77%	0.90%	2.45% / 1.26%	0.118

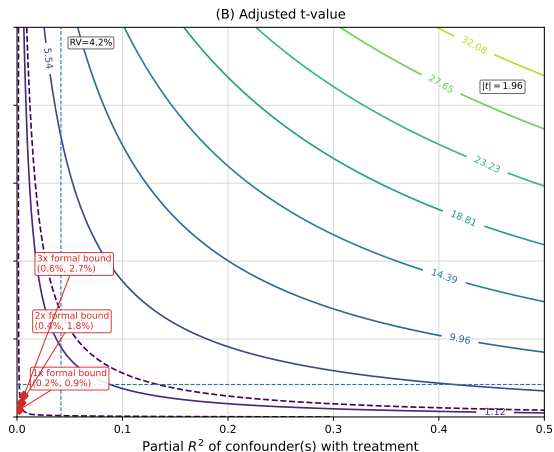
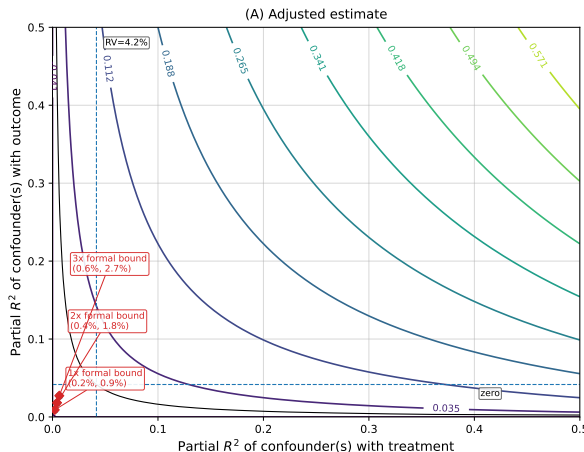
Notes: Steps are abbreviated as FE → Basic, Basic → Parent DV, FE → Full, and Basic → Full. Country blocks are ordered by the Step 2 restricted-model estimate in ascending order, so more negative coefficients appear first. The table reports Stata-weighted Cinelli–Hazlett sensitivity statistics. Specifications that add the household-background block use indicators for physical abuse by the respondent's mother/stepmother and father/stepfather and whether her father ever beat her mother. Eswatini did not collect the last item, so its specifications use only the two parental/stepparental physical-abuse indicators. $RV_{\alpha=.05}$ uses the robust t -statistic. The benchmark column reports the formal bound for a confounder as strong as the added observed-control block. The raw adjusted estimate is diagnostic only.

Summary, Range of All possible β_{long} 2

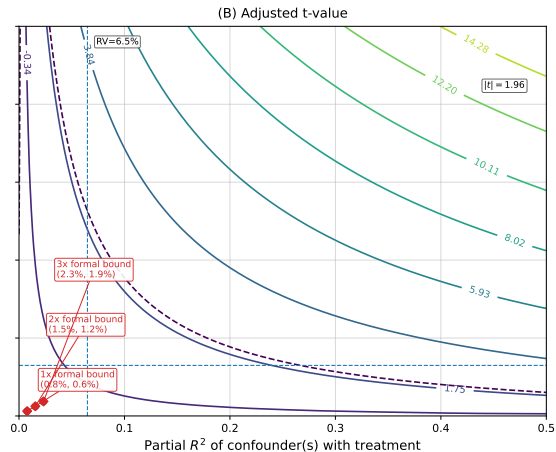
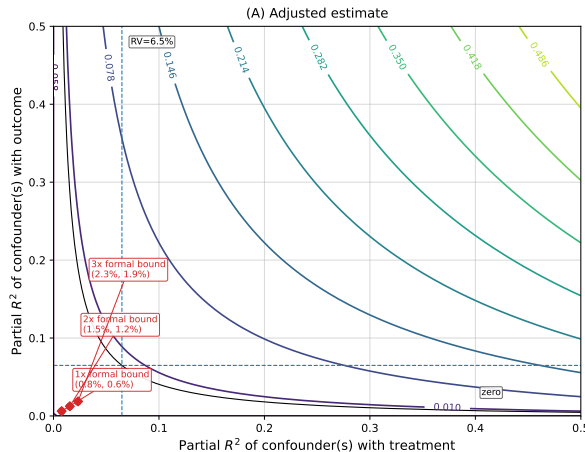


Note: Countries are ordered by the Step 2 restricted-model estimate (ascending). Eswatini did not collect whether the respondent's father ever beat her mother, so its controlled model uses only indicators of physical abuse by her mother/stepmother and father/stepfather.

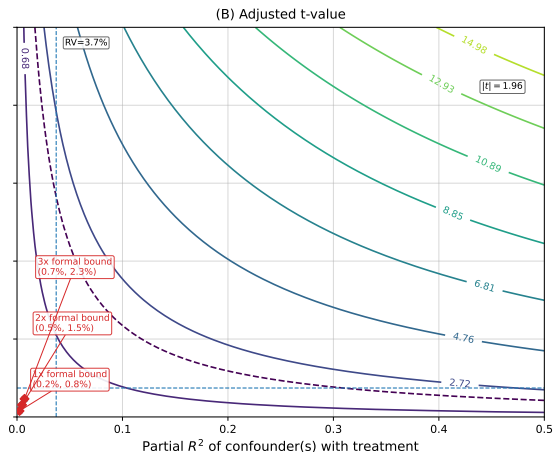
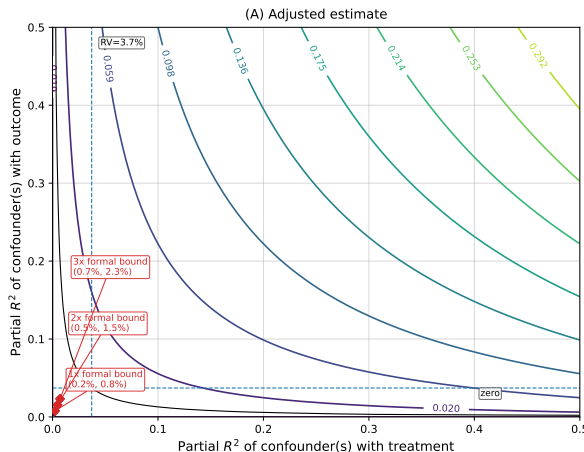
Benin, Range of All possible β_{long} 2



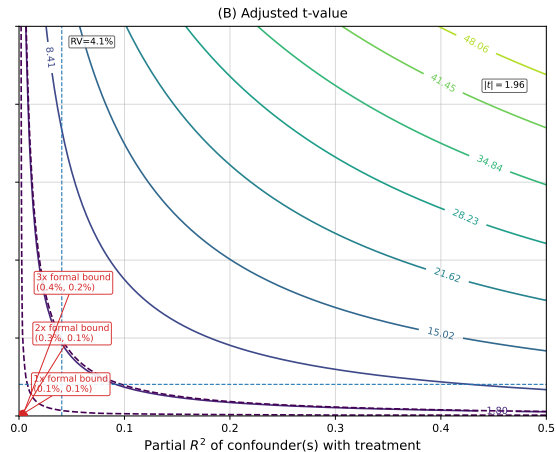
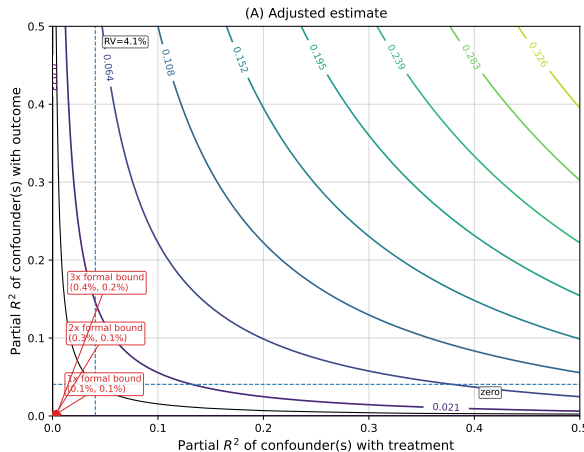
Comoros, Range of All possible β_{long} 2



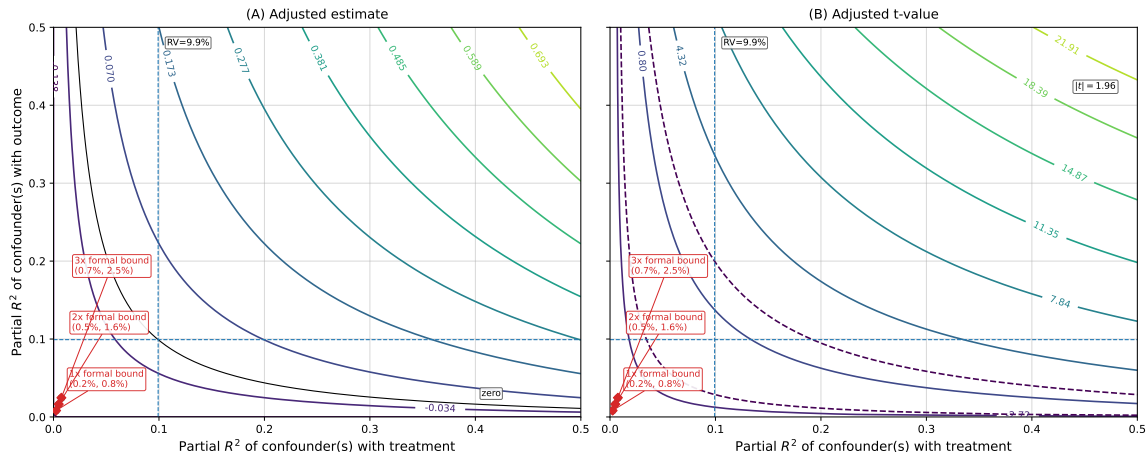
Eswatini, Range of All possible β_{long} 2



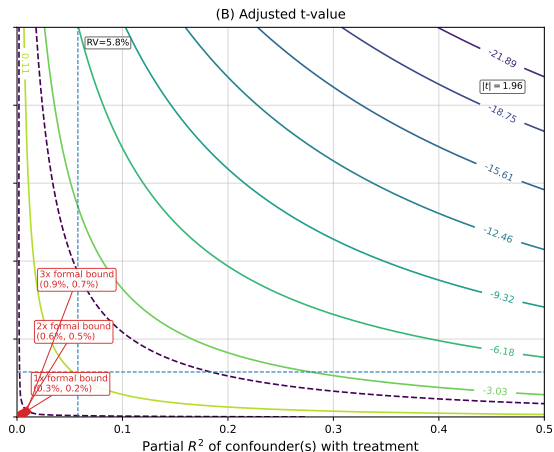
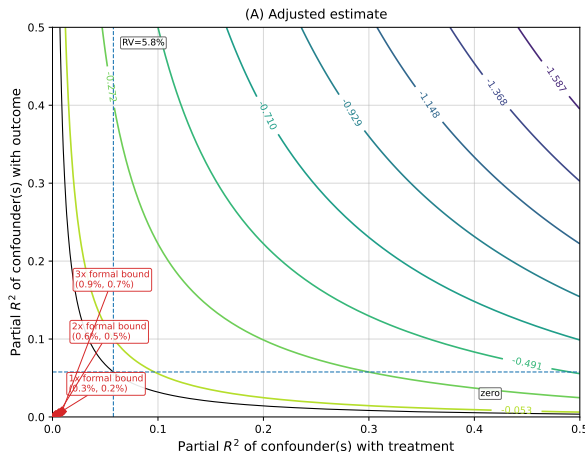
Honduras, Range of All possible β_{long} 2



Madagascar, Range of All possible β_{long} 2



Samoa, Range of All possible β_{long} 2



Tonga, Range of All possible β_{long} 2

